

# Algorithms on Lattices

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This document contains lecture notes from the course Algorithms on Lattices in the winter semester 2025/26, taught by Pavlo Yatsyna. The outline and content of the notes correspond directly to the content of the lectures.

Here is what will be on the exam:

- state the *structure theorem* for finitely generated modules over a principal ideal domain (14)
- define *HNF* or *SNF* + *existence and uniqueness* (algorithms not asked) (13, 14)
- there will be another question covering anything from what was in the course. These include definitions, statements of results, and their proofs. For example: Mikowski's 2nd theorem, SVP in 2D is polytime, complexity of SVP VS CVP, etc.

*This document was created with the help of Gemini and may contain mistakes.*

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# 1 Lattices and Discrete Groups

In this lecture, we introduce the concept of point lattices from two perspectives: a constructive definition using bases and an algebraic characterization involving discrete subgroups.

**Definition** (Lattice). Let  $\mathbb{R}^m$  be the  $m$ -dimensional Euclidean space. A lattice in  $\mathbb{R}^m$  is the set

$$\mathcal{L}(b_1, \dots, b_n) = \left\{ \sum_{i=1}^n x_i b_i : x_i \in \mathbb{Z} \right\}$$

of all integral linear combinations of  $n$  linearly independent vectors  $b_1, \dots, b_n$  in  $\mathbb{R}^m$  ( $m \geq n$ ). Using matrix notation, if  $B = [b_1, \dots, b_n] \in \mathbb{R}^{m \times n}$  is the basis matrix, we write:

$$\mathcal{L}(B) = \{Bx : x \in \mathbb{Z}^n\}.$$

**Definition** (Lattice basis). The sequence of vectors  $b_1, \dots, b_n$  is called a lattice basis.

**Definition** (Rank, dimension, full-rank). The integers  $n$  and  $m$  are called the rank and dimension of the lattice, respectively. If  $n = m$ , the lattice is called full rank.

## Algebraic Characterization.

Alternatively, lattices can be characterized without reference to a specific basis using the topological property of discreteness.

**Definition** (Discrete Subgroup). A subgroup  $\Gamma$  of a topological vector space  $V$  (e.g.,  $\mathbb{R}^m$ ) is called discrete if every point  $\gamma \in \Gamma$  is isolated, i.e., there exists a neighborhood of  $\gamma$  containing no other point of  $\Gamma$ .

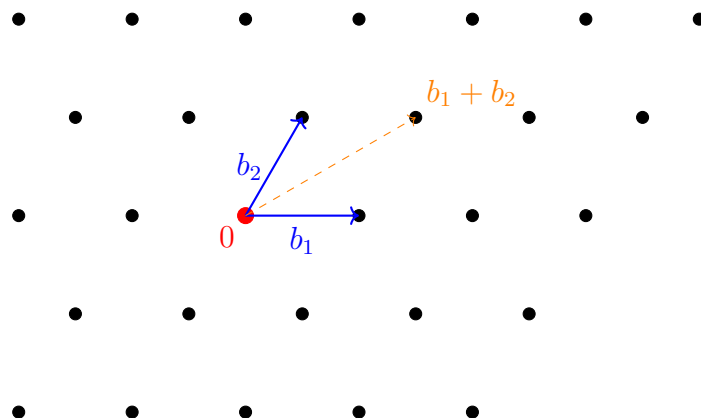
There is a fundamental equivalence between lattices and discrete subgroups in Euclidean space, as established in Minkowski's theory:

**Theorem** (Lattices as discrete subgroups). A subgroup  $\Gamma \subseteq \mathbb{R}^m$  is a lattice  $\iff$  it is discrete.

*Sketch.* ( $\Rightarrow$ ) A lattice generated by linearly independent vectors is clearly discrete. ( $\Leftarrow$ ) Conversely, if  $\Gamma$  is discrete, one can show it is closed. By selecting a maximal set of linearly independent vectors in  $\Gamma$  and considering the fundamental mesh, one proves that  $\Gamma$  must be finitely generated over  $\mathbb{Z}$ .  $\square$

## Visualization.

Graphically, a lattice looks like a regular grid of points. The following image illustrates a 2-dimensional lattice generated by basis vectors  $b_1$  and  $b_2$ .



## Bases of Lattices and Blichfeldt's Theorem

### Change of Basis and Unimodular Matrices

**Definition** (Unimodular matrix). A matrix  $U \in \mathbb{Z}^{n \times n}$  is called unimodular if  $\det(U) = \pm 1$ .

We denote  $I_n = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} \in M_n(\mathbb{Z})$  and  $GL_n(\mathbb{Z}) = \{\text{set of all } n \times n \text{ unimodular matrices}\}$ .

**Lemma.** Let  $A \in M_n(\mathbb{Z})$ . Then  $A^{-1} \in M_n(\mathbb{Z}) \iff A$  is unimodular.

*Proof.* We prove both implications.

“ $\Rightarrow$ ” Since  $AA^{-1} = I$ , we have  $\det(A)\det(A^{-1}) = 1$ . Both determinants are integers (as the matrices are integral), so  $\det(A)$  must be 1 or  $-1$ .

“ $\Leftarrow$ ” By Cramer's rule,  $A^{-1} = \det(A)^{-1} \cdot \text{adj}(A)$ . Since  $\det(A) = \pm 1$  and the adjugate matrix  $\text{adj}(A)$  consists of integer cofactors,  $A^{-1}$  is an integer matrix. □

**Proposition.** Let  $\mathcal{L} \subset \mathbb{R}^m$  be a lattice with basis  $V = (v_1, \dots, v_n)$ . Let  $W = (w_1, \dots, w_n)$  be vectors in  $\mathcal{L}$ . Then  $W$  is a basis of  $\mathcal{L} \iff$  there exists a unimodular matrix  $U \in M_n(\mathbb{Z})$  such that  $W = VU$ .

*Proof.* We prove both implications.

“ $\Rightarrow$ ” Since  $W$  is a basis,  $V \subset \mathcal{L}(W)$ , so  $V = WA$  for some  $A \in M_n(\mathbb{Z})$ . Similarly,  $W = VB$  for some  $B \in M_n(\mathbb{Z})$ . Thus  $V = VBA$ , implying  $BA = I$ . Hence  $B$  is unimodular.

“ $\Leftarrow$ ” If  $W = VU$  with  $U$  unimodular, then  $W \subset \mathcal{L}(V)$ . Also  $V = WU^{-1}$ . Since  $U^{-1} \in M_n(\mathbb{Z})$ ,  $V \subset \mathcal{L}(W)$ . Thus  $\mathcal{L}(V) = \mathcal{L}(W)$ . □

**Example.** Let  $\mathcal{L} = \mathbb{Z}^n$  with the standard basis  $E = (e_1, \dots, e_n) = I_n$ . A set of vectors  $V = (v_1, \dots, v_n)$  forms a basis of  $\mathbb{Z}^n \iff$  the matrix  $V$  is unimodular.

*Proof.* Using the proposition above:  $V$  is a basis  $\iff V = EU = IU = U$  for some unimodular matrix  $U$ . Thus  $V$  must be unimodular. □

### Determinant and Orthogonalization.

**Definition** (Full-rank lattice). Lattice  $\mathcal{L} \subseteq \mathbb{R}^n$  is full-rank ( $n = m$ ) if it is generated by  $n$  linearly independent vectors.

**Definition** (Determinant). Let  $\mathcal{L}$  be a full-rank lattice ( $n = m$ ). The determinant of the lattice is defined as  $\det(\mathcal{L}) = |\det(V)|$ , where  $V$  is any basis of  $\mathcal{L}$ . Also  $\det(\mathcal{L}(B)) = \text{vol}(\Phi(B)) = \prod \|b_i^*\|$ .

**Remark.** The determinant is independent of the choice of basis. If  $W = VU$  is another basis (with  $U$  unimodular), then:

$$|\det(W)| = |\det(VU)| = |\det(V)| \cdot |\det(U)| = |\det(V)| \cdot 1.$$

Elementary column operations preserve the lattice:

- (1) Swapping  $v_i$  and  $v_j$ .
- (2) Multiplying  $v_i$  by  $-1$ .
- (3) Add a basis vector to each other  $v_i \leftarrow v_i + cv_j$ .

**Gram-Schmidt Orthogonalization.** A possible way to compute the determinant. For any vectors  $\mathbf{b}_1, \dots, \mathbf{b}_n$ , define the corresponding Gram-Schmidt orthogonalized vectors  $\mathbf{b}_1^*, \dots, \mathbf{b}_n^*$  by

$$\mathbf{b}_i^* = \mathbf{b}_i - \sum_{j=1}^{i-1} \mu_{i,j} \mathbf{b}_j^*, \quad \mu_{i,j} = \frac{\langle \mathbf{b}_i, \mathbf{b}_j^* \rangle}{\langle \mathbf{b}_j^*, \mathbf{b}_j^* \rangle},$$

where  $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^m x_i y_i$  is the inner product in  $\mathbb{R}^m$ . For every  $i$ ,  $\mathbf{b}_i^*$  is the component of  $\mathbf{b}_i$  orthogonal to  $\mathbf{b}_1, \dots, \mathbf{b}_{i-1}$ . In particular,  $\text{span}(\mathbf{b}_1, \dots, \mathbf{b}_i) = \text{span}(\mathbf{b}_1^*, \dots, \mathbf{b}_i^*)$  and vectors  $\mathbf{b}_i^*$  are pairwise orthogonal, i.e.,  $\langle \mathbf{b}_i^*, \mathbf{b}_j^* \rangle = 0$  for all  $i \neq j$ . The determinant of the lattice equals the product of the lengths of the orthogonalized vectors

$$\det(\mathcal{L}(\mathbf{B})) = \prod_{i=1}^n \|\mathbf{b}_i^*\|.$$

## Successive Minima.

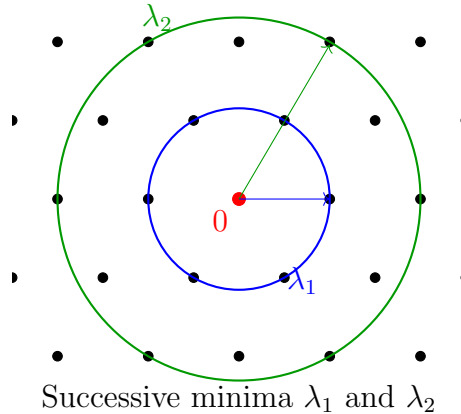
Let  $\lambda_1(\mathcal{L})$  denote the length of the shortest non-zero vector in  $\mathcal{L}$ .

$$\lambda_1(\mathcal{L}) = \min\{\|v\| : v \in \mathcal{L} \setminus \{0\}\}.$$

**Remark.** Using Gram-Schmidt, one can show  $\lambda_1(\mathcal{L}) \geq \min_i \|\mathbf{v}_i^*\|$ .

**Definition** (Successive Minima). The  $i$ -th successive minimum  $\lambda_i(\mathcal{L})$  is the smallest radius  $r$  such that the closed ball  $\bar{B}(0, r)$  contains at least  $i$  linearly independent lattice vectors.

$$\lambda_i = \lambda_i(\mathcal{L}) = \inf\{r \mid \dim(\text{span}(\mathcal{L} \cap B(0, r))) \geq i\}.$$



## Geometry of Numbers.

**Theorem.** Let  $\mathcal{L}$  be a full-rank lattice of dimension  $n$ . Then there exist linear independent vectors  $v_1, \dots, v_n \in \mathcal{L}$  such that  $\|v_i\| = \lambda_i$  for  $i = 1, \dots, m$ .

**Motivation.** We want to show that if a set is "large enough" relative to the lattice determinant, it must contain two points whose difference is in the lattice. This is a generalization of the Pigeonhole Principle.

**Theorem** (Blichfeldt). Let  $\mathcal{L} \subset \mathbb{R}^n$  be a full-rank lattice and  $S \subset \mathbb{R}^n$  be a measurable set. If  $\text{vol}(S) > \text{vol}(\mathcal{L})$ , then there exist vectors  $x, y \in S, x \neq y$  such that  $x - y \in \mathcal{L}$ .

*Proof.* Let  $\Phi = \{y_1 v_1 + \dots + y_n v_n \mid y \in [0, 1]\}$  be the parallelepiped of the lattice basis  $V$ . We know  $\text{vol}(\Phi) = \det(\mathcal{L})$ . Partition  $S$  (subset of  $\text{span}(\mathcal{L})$ ) according to the lattice cells:

$$S = \bigcup_{v \in \mathcal{L}} (S \cap (\Phi + v)).$$

Let  $S_v = S \cap (\Phi + v)$ . We translate these pieces back to the fundamental domain by defining  $S'_v = S_v - v \subseteq \Phi$ . Since translation preserves volume:

$$\text{vol}(S) = \sum_{v \in \mathcal{L}} \text{vol}(S_v) = \sum_{v \in \mathcal{L}} \text{vol}(S'_v).$$

By assumption,  $\sum \text{vol}(S'_v) > \text{vol}(\Phi)$ . Since all  $S'_v$  lie inside  $\Phi$ , they cannot be disjoint (by measure theory/pigeonhole principle). Thus, there exist  $v_1 \neq v_2$  and a point  $z \in S'_{v_1} \cap S'_{v_2}$ . By definition of  $S'_v$ , there exist  $x \in S_{v_1} \subseteq S$  and  $y \in S_{v_2} \subseteq S$  such that:

$$z = x - v_1 = y - v_2.$$

Hence  $x - y = v_1 - v_2$ . Since  $v_1, v_2 \in \mathcal{L}$ , their difference is in  $\mathcal{L}$ . Since  $v_1 \neq v_2$ ,  $x \neq y$ . □

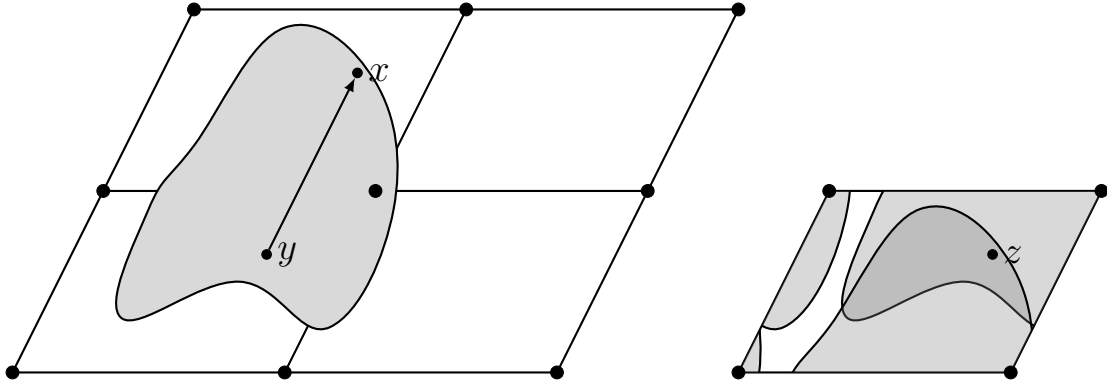


Figure: Blichfeldt's theorem.

### 3 Minkowski's Theorems and Lattice Problems

**Definition** (Centrally symmetric set). A set  $S \subseteq \mathbb{R}^n$  is called centrally symmetric if  $\forall x \in S$ , it holds that  $-x \in S$ .

Geometrically, this means the set is symmetric with respect to the origin. Examples include balls centered at 0, hypercubes centered at 0, etc.

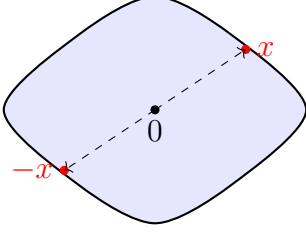


Figure 1: Centrally Symmetric Set.

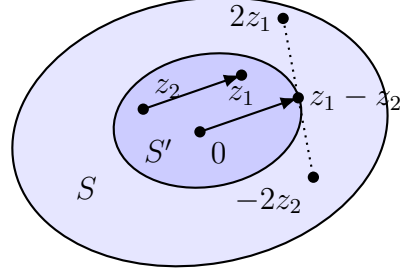


Figure 2: Minkowski's convex body Theorem.

Minkowski's first theorem is a fundamental result in the Geometry of Numbers, relating the volume of a set to the existence of lattice points.

**Theorem** (Minkowski's convex body). Let  $\mathcal{L}$  be a full-rank lattice in  $\mathbb{R}^n$ . Let  $S \subset \mathbb{R}^n$  be a convex, centrally symmetric set with volume

$$\text{vol}(S) > 2^n \det(\mathcal{L}).$$

Then  $S$  contains at least one non-zero lattice point (i.e.,  $S \cap (\mathcal{L} \setminus \{0\}) \neq \emptyset$ ).

*Proof.* Consider the set  $S' = \frac{1}{2}S = \{\frac{1}{2}x : x \in S\}$ . The volume scales with the dimension  $n$ , so:

$$\text{vol}(S') = \left(\frac{1}{2}\right)^n \text{vol}(S) = \frac{1}{2^n} \text{vol}(S).$$

By the assumption  $\text{vol}(S) > 2^n \det(\mathcal{L})$ , we have  $\text{vol}(S') > \det(\mathcal{L})$ .

By Blichfeldt's theorem, since  $\text{vol}(S') > \det(\mathcal{L})$ , there exist points  $z_1, z_2 \in S'$ ,  $z_1 \neq z_2$  such that  $z_1 - z_2 \in \mathcal{L}$ . Since  $z_1, z_2 \in S'$ , we can write them as  $z_1 = \frac{1}{2}x$  and  $z_2 = \frac{1}{2}y$  for some  $x, y \in S$ . The difference is:

$$v = z_1 - z_2 = \frac{1}{2}x - \frac{1}{2}y.$$

Since  $S$  is centrally symmetric,  $y \in S \implies -y \in S$ . Since  $S$  is convex, we can take  $\theta = \frac{1}{2}$ :

$$\theta x + (1 - \theta)(-y) = \frac{1}{2}x + (1 - \frac{1}{2})(-y) = \frac{x - y}{2} = v.$$

Thus,  $v \in S$ . Since  $z_1 \neq z_2$ ,  $v \neq 0$ , also  $v \in \mathcal{L}$  by Blichfeldt. So  $v$  is a non-zero lattice point in  $S$ .  $\square$

We can apply Minkowski's theorem to bound the length of the shortest non-zero vector,  $\lambda_1(\mathcal{L})$ .

**Corollary.** For any full-rank lattice  $\mathcal{L} \subseteq \mathbb{R}^n$ , there exists a non-zero vector  $v \in \mathcal{L}$  such that:

$$\|v\| \leq \sqrt{n} \det(\mathcal{L})^{1/n}.$$

*Proof.* Let  $B = \{v \in \mathbb{R}^n \mid \|v\| < \sqrt{n} \det(\mathcal{L})^{1/n}\}$ . We want to apply Minkowski theorem above, so we need to show that  $B$  is convex, centrally-symmetric and has  $\text{vol}(B) > 2^n \det(\mathcal{L})$ . Let  $S$  be the cube centered at the origin with side length  $2B$ :

$$S = \{x \in \mathbb{R}^n : \|x\|_\infty < B\}.$$

Note that for any  $x \in S$ , the Euclidean norm satisfies  $\|x\|_2 < \sqrt{n}B$ . The volume is  $\text{vol}(S) = (2B)^n = 2^n B^n$ . We want  $\text{vol}(S) > 2^n \det(\mathcal{L})$ , which implies:

$$2^n B^n > 2^n \det(\mathcal{L}) \iff B^n > \det(\mathcal{L}) \iff B > \det(\mathcal{L})^{1/n}.$$

Let us choose  $B = \det(\mathcal{L})^{1/n} + \epsilon$  for arbitrary small  $\epsilon$ . By *Minkowski's theorem*, there exists a non-zero  $v \in S \cap \mathcal{L}$ . Since  $v \in S$ , we have  $\|v\|_\infty < B$ . In terms of Euclidean norm:

$$\|v\|_2 \leq \sqrt{n}\|v\|_\infty < \sqrt{n}B.$$

Taking the limit  $\epsilon \rightarrow 0$ , we get  $\|v\|_2 \leq \sqrt{n} \det(\mathcal{L})^{1/n}$ . □

The following theorem relates the geometric mean of the successive minima to the determinant.

**Theorem** (Minkowski's Second Theorem). *Let  $\mathcal{L}$  be a full-rank lattice in  $\mathbb{R}^n$  with successive minima  $\lambda_1, \dots, \lambda_n$ . Then:*

$$\left( \prod_{i=1}^n \lambda_i \right)^{1/n} < \sqrt{n} \det(\mathcal{L})^{1/n}.$$

*Proof.* Let  $x_1, \dots, x_n \in \mathcal{L}$  be linearly independent lattice vectors achieving the successive minima, i.e.,  $\|x_i\| = \lambda_i$ . Assume for the sake of contradiction that:

$$\prod_{i=1}^n \lambda_i \geq (\sqrt{n})^n \det(\mathcal{L}).$$

Consider the Gram-Schmidt orthogonalized vectors  $x_i^*$  derived from the sequence  $x_1, \dots, x_n$ . Define a linear transformation  $T$  that expands the space along the axes defined by  $x_i^*$  by a factor of  $\lambda_i$ :

$$T\left(\sum c_i x_i^*\right) = \sum \lambda_i c_i x_i^*.$$

Let  $S = \mathcal{B}(0, 1) \cap \text{span}(\mathcal{L})$  be the  $n$ -dimensional open unit ball. Applying  $T$  to  $S$  yields a centrally symmetric convex body  $T(S)$ . The volume of this body is:

$$\text{vol}(T(S)) = \left( \prod_{i=1}^n \lambda_i \right) \text{vol}(S).$$

Using our assumption  $\prod \lambda_i \geq (\sqrt{n})^n \det(\mathcal{L})$  and the fact that  $\text{vol}(\sqrt{n}S) > \text{vol}(C) = 2^n$  (ball of radius  $\sqrt{n}$  contains a hypercube  $C = \{x_i \in \mathbb{R}^n \mid \|x_i\| < 1\}$ ), we derive:

$$\text{vol}(T(S)) \geq \text{vol}(\sqrt{n}S) \det(\mathcal{L}) > 2^n \det(\mathcal{L}).$$

By *Minkowski's convex body Theorem*,  $T(S)$  must contain a non-zero lattice point  $y \in T(S) \cap \mathcal{L} \setminus \{0\}$ . Since  $y \in T(S)$ , it must be the image of some point  $x \in S$ , i.e.,  $y = T(x)$  with  $\|x\| < 1$ . We can express  $x$  and  $y$  in the basis  $x_i^*$ :

$$x = \sum_{i=1}^n c_i x_i^*, \quad y = \sum_{i=1}^n \lambda_i c_i x_i^*.$$

Since  $y \neq 0$ , some coefficient  $c_i$  is non-zero. Let  $k$  be the largest index such that  $c_k \neq 0$ . Let  $k' \leq k$  be the smallest index such that  $\lambda_{k'} = \lambda_k$  (handling the case where multiple minima are equal).

First, we observe that  $y$  is linearly independent from  $x_1, \dots, x_{k'-1}$ . This is because the inner product  $\langle y, x_k^* \rangle = \lambda_k c_k \|x_k^*\|^2 \neq 0$ , whereas  $x_k^*$  is orthogonal to the  $\text{span}(x_1, \dots, x_{k'-1})$  (which is the same as the  $\text{span}(x_1^*, \dots, x_{k'-1}^*)$ ).

Second, we show that the length of  $y$  is strictly less than  $\lambda_k$ :

$$\|y\|^2 = \left\| \sum_{i=1}^k \lambda_i c_i x_i^* \right\|^2 = \sum_{i=1}^k \lambda_i^2 c_i^2 \|x_i^*\|^2.$$

Since  $\lambda_1 \leq \dots \leq \lambda_k$ , we have  $\lambda_i \leq \lambda_k$  for all  $i \leq k$ . Thus:

$$\|y\|^2 \leq \sum_{i=1}^k \lambda_k^2 c_i^2 \|x_i^*\|^2 = \lambda_k^2 \left\| \sum_{i=1}^k c_i x_i^* \right\|^2 = \lambda_k^2 \|x\|^2.$$

Since  $x \in S$ , we know  $\|x\| < 1$ , and therefore  $\|y\| < \lambda_k$ .

We have found a non-zero lattice vector  $y$  that is linearly independent of  $x_1, \dots, x_{k'-1}$  and has length strictly less than  $\lambda_k = \lambda_{k'}$ . This implies there are  $k'$  linearly independent lattice vectors  $x_1, \dots, x_{k'-1}, y$  with lengths strictly less than  $\lambda_{k'}$ . This contradicts the definition of the  $k'$ -th successive minimum.  $\square$



## 4 SVP Variants and Lattices in Dimension 2

The Shortest Vector Problem (SVP) is central to the field of lattice-based cryptography. Their importance is twofold: on the *constructive* side, the assumed hardness of solving SVP in high dimensions underpins the security of modern encryption schemes that are believed to be resistant to quantum computers. On the *destructive* side, efficient algorithms that approximate SVP are powerful tools for cryptanalysis, used to break systems like knapsack cryptosystems or specific RSA implementations.

### SVP variants.

We formally define the variants of SVP. Let  $\mathcal{L} \subseteq \mathbb{R}^n$  be a lattice given (input) by a basis  $B$ .

- **SVP (Search):** Find  $v \in \mathcal{L} \setminus \{0\}$  such that  $\|v\| = \lambda_1(\mathcal{L})$ .
- **SVP (Optimization):** Find  $\lambda_1(\mathcal{L})$ .
- **SVP (Decision):** Given approximation factor  $\gamma : \mathbb{N} \rightarrow \mathbb{R}$ , decide if  $\lambda_1(\mathcal{L}) \leq \gamma$ .

For approximation factors  $\gamma \geq 1$  we define:

- **SVP $_\gamma$  (Search):** Find  $v \in \mathcal{L} \setminus \{0\}$  such that  $\|v\| \leq \gamma \cdot \lambda_1(\mathcal{L})$ .
- **SVP $_\gamma$  (Optimization):** Find  $d \in \mathbb{Q}^+$  such that  $\lambda_1(\mathcal{L}) \leq d \leq \gamma \cdot \lambda_1(\mathcal{L})$ .
- **GapSVP $_\gamma$  (Promise Problem):** Given pair  $r \in \mathbb{Q}^+$ , decide between:
  - YES:  $\lambda_1(\mathcal{L}) \leq r$
  - NO:  $\lambda_1(\mathcal{L}) > \gamma \cdot r$

### Lattices in Dimension 2.

In general, we care about *reduced bases* rather than just *shortest* because finding the exact shortest vector is NP-hard in high dimensions. A reduced basis offers a trade-off: it can be polytime and provides *short and orthogonal enough* vectors to solve many problems.

We now focus on solving SVP exactly in dimension  $n = 2$ . Let  $\mathcal{L}$  be generated by a basis  $(a, b)$ .

**Definition** (Shortest Basis). *We say that  $(a, b)$  a shortest basis of a full-rank lattice  $\mathcal{L}$ , if:*

- (1)  $a, b$  is a basis of  $\mathcal{L}$ ,
- (2)  $\|a\| = \lambda_1(\mathcal{L})$ ,
- (3)  $\|b\| = \lambda_2(\mathcal{L})$ , i.e.  $\forall v \in \mathcal{L} \setminus \langle a \rangle_{\mathbb{R}} : \|v\| \geq \|b\|$ .

**Definition** (Reduced Basis). *A basis  $(a, b)$  is called reduced if  $\|a\|, \|b\| \leq \|a + b\|$ .*

**Lemma.** *Let  $x, y \in \mathbb{R}^n$  and  $\alpha \in (1, \infty)$ . If  $\|x\| < \|x + y\|$ , then  $\|x + y\| < \|x + \alpha y\|$ .<sup>1</sup>*

*Proof.* Let  $c = \frac{1}{\alpha} > 0$ . We can rewrite  $x + y$  as follows:

$$x + y = x(-cx + cx) + (c \cdot \alpha)y = (1 - c)x + c(x + \alpha y).$$

Now, by triangular inequality (\*) and by assumption  $\|x\| < \|x + y\|$  (\*\*), we get

$$\begin{aligned} \|x + y\| &\stackrel{*}{\leq} (1 - c)\|x\| + c\|x + \alpha y\| \stackrel{**}{<} (1 - c)\|x + y\| + c\|x + \alpha y\| \\ \|x + y\| &< \|x + y\| - c\|x + y\| + c\|x + \alpha y\| \\ c\|x + y\| &< c\|x + \alpha y\|, \quad \text{and for } c > 0 : \quad \|x + y\| < \|x + \alpha y\| \end{aligned}$$

□

---

<sup>1</sup>Holds also for “ $\leq$ ”.

**Theorem.** Let  $(a, b)$  be a basis of a full-rank lattice  $\mathcal{L} \subset \mathbb{R}^2$ . Then  $(a, b)$  is reduced  $\iff (a, b)$  is a shortest basis.

*Proof.* We prove both implications.

“ $\Leftarrow$ ” Assume  $a, b$  is a shortest basis with  $\|a\| \leq \|b\|$ . By definition,  $\|a\| = \lambda_1(\mathcal{L})$  and  $\|b\| = \lambda_2(\mathcal{L})$ . Consider the vector  $a + b$ . Since  $a, b$  is a basis,  $a$  and  $b$  are linearly independent, thus  $a + b \neq 0$ . Since  $a + b$  is linearly independent of  $a$ , it belongs to  $\mathcal{L} \setminus \langle a \rangle_{\mathbb{R}}$ . By the definition of  $\lambda_2$ :

$$\|a + b\| \geq \lambda_2(\mathcal{L}) = \|b\| \geq \|a\|.$$

Thus  $\|a\|, \|b\| \leq \|a + b\|$ , so the basis is reduced.

“ $\Rightarrow$ ” Assume  $a, b$  is reduced, i.e.,  $\|a\| \leq \|b\|$  and  $\|a\|, \|b\| \leq \|a + b\|$ . We want to show that  $\|a\| = \lambda_1(\mathcal{L})$  and  $\|b\| = \lambda_2(\mathcal{L})$ . Let  $v \in \mathcal{L} \setminus \{0\}$ . We can write  $v = ra + sb$  for  $r, s \in \mathbb{Z}$ . We analyze the norm  $\|v\|$  in three cases:

**Case 1:**  $s = 0$ . Then  $v = ra$ . Since  $v \neq 0$ , we have  $r \neq 0$ , so  $|r| \geq 1$ .

$$\|v\| = |r|\|a\| \geq \|a\|.$$

This shows that  $a$  is indeed the shortest non-zero vector, so  $\|a\| = \lambda_1(\mathcal{L})$ .

**Case 2:**  $r = 0$ . Then  $v = sb$ . Since  $s \neq 0$ ,  $|s| \geq 1$ .

$$\|v\| = |s|\|b\| \geq \|b\|.$$

**Case 3:**  $r \neq 0$  and  $s \neq 0$ . Assume  $r \geq s \geq 0$ . And since  $s \in \mathbb{Z} \setminus \{0\}$ ,  $|s| \geq 1$ . We need to show  $\|ra + sb\| \geq \|b\|$ . By reduction, it holds that  $\|a\|, \|b\| \leq \|a + b\|$ . Hence, by the *lemma* above (\*), there must  $\exists \alpha \geq 1$  such that  $\|a + b\| \leq \|a + \alpha b\|$ . Since  $r \geq s \geq 0$  and  $|s| > 1$ , also  $\frac{r}{s} > 1$ , so let  $\alpha = \frac{r}{s}$ . Then we have

$$\|a\|, \|b\| \leq \|a + b\| \stackrel{*}{\leq} \|b + (\frac{r}{s})a\| = \|\frac{ra + sb}{s}\| = s\|ra + sb\| \stackrel{|s| \geq 1}{\leq} \|ra + sb\|.$$

□

## Gauss' Algorithm.

This algorithm (essentially the Euclidean algorithm for vectors) constructs a reduced basis.

**Definition.** A pair  $(a, b)$  is well-ordered if  $\|a\| \leq \|a - b\| \leq \|b\|$ .

**Algorithm** (Gauss).

*Input:* Basis  $a, b \in \mathbb{Z}^2$ . of  $\mathcal{L} \subseteq \mathbb{R}^2$ .

*Output:* Shortest basis of  $\mathcal{L}$ .

### 1. Init:

- If  $\|a\| > \|b\|$ , then swap  $a$  and  $b$ .
- If  $\|a - b\| > \|a + b\|$ , then  $b := -b$ . (*Ensure acute angle*)
- If  $\|a\| = \|b\|$ , return  $a, a - b$ .
- If  $\|b\| \leq \|a - b\|$ , return  $a, b$ .
- If  $\|a\| \leq \|a - b\|$ , goto Loop (step 2).
- Let  $a := b - a$ ,  $b := a$ .

### 2. Loop:

- Find  $\mu \in \mathbb{Z}$  such that  $\|b - \mu a\|$  is minimal (i.e.,  $\mu = \lfloor \langle b, a \rangle / \|a\|^2 \rfloor$ ).

- Update  $b := b - \mu a$ .
- If  $\|a - b\| > \|a + b\|$ , then  $b := -b$  and swap  $a, b$ .
- If  $a, b$  is reduced (i.e.,  $\|a\| \leq \|b\| \leq \|a + b\|$ ), return  $a, b$ .
- Else swap  $a, b$  (since now  $\|b\| < \|a\|$ ) and Goto Loop.

**Lemma.** *In each iteration of the Loop, the pair  $a, b$  is well-ordered at the beginning of the step.*

*Proof.* In the first iteration, this is guaranteed by the *Init* phase. In subsequent iterations, the loop repeats only if the condition "If  $a, b$  reduced" fails. If it fails due to order (meaning after reduction  $\|b_{new}\| < \|a\|$ ), the algorithm performs a swap. Thus, at the start of the next iteration (where  $a$  is the old  $b_{new}$ ), the condition  $\|a\| \leq \|b\|$  holds.  $\square$

**Theorem.** *The Gauss algorithm terminates and produces a reduced basis of  $\mathcal{L}$ .*

*Proof.* We prove termination and correctness.

- *Termination:* Consider the sequence of vectors assigned to  $a$ . In every step where the loop repeats (a swap occurs), the new vector  $a$  is the previous vector  $b$  (which was reduced by  $\mu a$ ). Since a swap only happens if  $\|b_{reduced}\| < \|a\|$ , the length of the vector  $a$  strictly decreases in each iteration:  $\|a_{i+1}\| < \|a_i\|$ . Since  $\mathcal{L}$  is a discrete lattice, there is a minimum possible length for non-zero vectors ( $\lambda_1$ ). A sequence of strictly decreasing discrete values must be finite. Thus, the algorithm terminates.
- *Correctness:* The algorithm returns only when the "If  $a, b$  reduced" condition is met.
  - (1) The step  $b := b - \mu a$  with  $\mu = \lfloor \langle b, a \rangle / \|a\|^2 \rfloor$  ensures that the projection coefficient satisfies  $|\mu| \leq 1/2$ .
  - (2) The check  $\|a - b\| > \|a + b\|$  ensures the angle is not obtuse (consistent with  $\|b\| \leq \|a + b\|$ ).
  - (3) The condition  $\|a\| \leq \|b\|$  is checked before returning.

Combined, these ensure the output satisfies the definition of a reduced basis.  $\square$

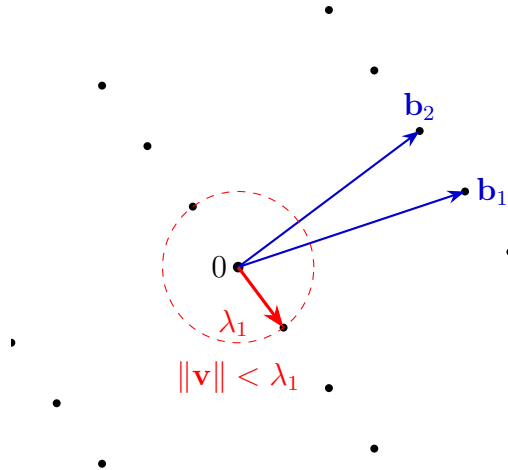


Figure 1: *Example of SVP in  $\mathbb{R}^2$ .*

*This visualizes the "bad basis" scenario. We are given long basis vectors  $(b_1, b_2)$ , but the shortest vector  $\lambda_1$  is actually a combination of them.*

## 5 LLL-Reduced Basis

### Analysis of Gauss' Algorithm (Dimension 2)

We analyze the running time of the Gauss algorithm (Lagrange reduction) presented in the previous lecture. Let  $(a_1, a_2, \dots)$  be the sequence of vectors produced by the algorithm, where  $a_k$  represents the vector with the smaller norm at the beginning of the  $k$ -th iteration (after the swap, if any).

**Lemma** (Geometric Decay). *For  $i \geq 3$ , the norms satisfy:*

$$\|a_i\| < \frac{1}{2}\|a_{i+1}\|.$$

*Proof.* Let  $a_{i-1}, a_i, a_{i+1} = u, v, w$ . Let  $u, v$  and  $v, w$  be well-ordered pairs occurring in the algorithm and  $\|u\| < \|v\| < \|w\|$ . Claim

$$\|w\| > 2\|v\|.$$

The relationship given by the reduction step is  $u = \pm(w - \mu v)$  (since  $u$  is the remainder of  $w$  reduced by  $v$ ). Rewriting this for the vector  $w$ :

$$w = \varepsilon u + \mu v$$

where  $\varepsilon \in \{1, -1\}$  and  $\mu$  is the integer coefficient used in the reduction. We analyze the norm  $\|w\|$  based on the value of  $\mu$ . Note that since  $v, w$  were well-ordered before reduction,  $\mu \geq 1$  (otherwise no reduction/swap would occur).

- **Case 1:**  $\mu = 1$ . This is impossible at  $\|w - \mu v\| = \|w - v\| = \|u\| < \|v\|$ . That contradicts with  $v, w$  being well-ordered ( $\|v\| \leq \|v - w\| \leq \|w\|$ ).
- **Case 2:**  $\mu = 2$  and  $\varepsilon = -1$ . We have

$$w = 2v - u \iff w = v + v - u \iff w - v = v - u \iff \|w - v\| = \|v - u\|.$$

But this would contradict that either  $\|u - v\| < \|v\|$  or  $\|v\| < \|v - w\|$ .

- **Case 3:**  $\mu \geq 2$  and  $\varepsilon = 1$ . We know that  $\|v - u\| < \|v\|$  as  $u, v$  is well ordered. No, we use *lemma* from the previous lecture by taking  $x = v - u, y = u$  and  $\lambda = 2$ :

$$\|x\| \leq \|x + y\| \implies \|x + y\| \leq \|x + \lambda y\| \iff \|v - u\| \leq \|v\| \implies \|v\| \leq \|v + u\|$$

So we have  $\|u\| < \|v\| < \|u + v\|$ . Now we apply *lemma* again with  $\mu$ , i.e.  $\|u\| \leq \|u + \mu v\|$ . This gives that  $w = u + \mu v$  satisfy  $\|w\| = \|u + \mu v\| \geq \|2v + u\|$ . We want to show that  $\|2v + u\| > 2\|v\|$ .

Consider  $2v - u$ . By the triangular inequality and using the fact that  $\|u - v\| < \|v\|$  we get that

$$\|2v - u\| \leq \|v\| + \|v - u\| < 2\|v\|.$$

Apply *lemma* again by taking  $x = 2v - u$  and  $y = u$ :

$$\|2v - u\| < \|2v\| < \|2v + u\| \implies \|w\| > 2\|v\|.$$

□

We showed that for  $i \geq 3$ :  $\|a_i\| \geq 2^{i-3}\|a_3\|$ . For  $a, b$  we have that  $\|a\| + \|b\| \geq \|a_{k+1}\| \geq 2^{k-2}\|a_3\| \geq 2^{k-2}$ . Therefore  $k \leq 2 + \log_2(\|a\| + \|b\|)$ .

**Theorem.** *There exists a polynomial-time algorithm that, given two linearly independent vectors in  $\mathbb{R}^2$  generating a lattice  $\mathcal{L}$ , outputs a basis  $(a, b)$  of  $\mathcal{L}$  such that  $\|a\| = \lambda_1(\mathcal{L})$  and  $\|b\| = \lambda_2(\mathcal{L})$ .*

## Approximating SVP in Dimension $n$ .

We generalize the notion of a reduced basis to dimension  $n$ .

**Definition** (Reduced basis in  $\mathbb{R}^2$ ). A basis  $(a, b)$  is reduced if  $\mu = \frac{ab}{\|a\|^2} \leq \frac{1}{2}$  and  $\|a\| \leq \|b\|$ .

**Definition** (LLL-Reduced Basis). A basis  $B = (b_1, \dots, b_n)$  of a lattice  $\mathcal{L}$  is called LLL-reduced with parameter  $\delta = 3/4$  if:

(R1)  $|\mu_{i,j}| \leq \frac{1}{2}$ , for all  $1 \leq j < i \leq n$ :

(R2) For all  $1 < i \leq n$ :

$$\|b_i^* + \mu_{i,i-1}b_{i-1}^*\|^2 \geq \frac{3}{4}\|b_{i-1}^*\|^2.$$

**Remark.** The Lovász condition (R2) relates the length of the projection of  $b_i$  onto the orthogonal complement of  $\text{span}(b_1, \dots, b_{i-2})$  to the length of  $b_{i-1}^*$ . Using the Gram-Schmidt orthogonalization property, the LHS is:

$$\|b_i^* + \mu_{i,i-1}b_{i-1}^*\|^2 = \|b_i^*\|^2 + \mu_{i,i-1}^2\|b_{i-1}^*\|^2.$$

Thus, R2 implies:

$$\|b_i^*\|^2 \geq \left(\frac{3}{4} - \mu_{i,i-1}^2\right)\|b_{i-1}^*\|^2.$$

Since  $|\mu_{i,i-1}| \leq 1/2$  (by R1), we have  $\mu^2 \leq 1/4$ , so  $\|b_i^*\|^2 \geq \frac{1}{2}\|b_{i-1}^*\|^2$ .

## Properties of LLL Reduced Bases

**Lemma.** Let  $b_1, \dots, b_n$  be an LLL-reduced basis. Then for any  $1 \leq i \leq j \leq n$ :

$$\|b_i\|^2 \leq 2^{j-1}\|b_j^*\|^2.$$

*Proof.* From the Remark above,  $\|b_{k-1}^*\|^2 \leq 2\|b_k^*\|^2$ . By induction, for  $i \leq j$ :

$$\|b_i^*\|^2 \leq 2^{j-i}\|b_j^*\|^2.$$

Now consider the vector  $b_i$ . By Gram-Schmidt decomposition:

$$\|b_i\|^2 = \|b_i^*\|^2 + \sum_{k=1}^{i-1} \mu_{i,k}^2 \|b_k^*\|^2.$$

Using  $|\mu_{i,k}| \leq 1/2$  and the bound on  $\|b_k^*\|^2$ :

$$\|b_i\|^2 \leq \|b_i^*\|^2 + \sum_{k=1}^{i-1} \frac{1}{4} 2^{i-k} \|b_i^*\|^2 = \|b_i^*\|^2 \left(1 + \frac{1}{4} \sum_{p=1}^{i-1} 2^p\right) \leq \|b_i^*\|^2 \left(1 + \frac{1}{4} \cdot 2^{i-1}\right) \leq 2^{j-i+i-1} \|b_j^*\|^2.$$

Where we used the fact that  $\|b_i^*\|^2 \leq 2^{j-i}\|b_j^*\|^2$ . □

**Proposition** (Bounds on Basis Lengths). Let  $b_1, \dots, b_n$  be an LLL-reduced basis of  $\mathcal{L}$ . Then:

$$\det(\mathcal{L}) \leq \prod_{i=1}^n \|b_i\| \leq 2^{\frac{n(n-1)}{4}} \det(\mathcal{L}) \quad \text{and} \quad \|b_i\| \leq 2^{\frac{n-1}{2}} \sqrt[n]{\det(\mathcal{L})}.$$

*Proof.* We prove both bounds.

- **Lower Bound:** This is a direct application of *Hadamard's Inequality*. For any basis matrix  $B = [b_1, \dots, b_n]$ , we have:

$$\det(\mathcal{L}) = |\det(B)| \leq \prod_{i=1}^n \|b_i\|.$$

- **Upper Bound:** We use the *Lemma* above. Setting  $i = j$ , we have:

$$\|b_i\|^2 \leq 2^{i-1} \|b_i^*\|^2 \implies \|b_i\| \leq 2^{(i-1)/2} \|b_i^*\|.$$

Now, take the product over all  $i = 1, \dots, n$ :

$$\prod_{i=1}^n \|b_i\| \leq \prod_{i=1}^n (2^{(i-1)/2} \|b_i^*\|).$$

We can separate the powers of 2 and the norms of Gram-Schmidt vectors:

$$\prod_{i=1}^n \|b_i\| \leq \left( \prod_{i=1}^n 2^{(i-1)/2} \right) \left( \prod_{i=1}^n \|b_i^*\| \right).$$

The determinant of the lattice is the product of the Gram-Schmidt lengths:  $\det(\mathcal{L}) = \prod \|b_i^*\|$ . Calculating the exponent of 2:

$$\sum_{i=1}^n \frac{i-1}{2} = \frac{1}{2} \sum_{k=0}^{n-1} k = \frac{1}{2} \frac{n(n-1)}{2} = \frac{n(n-1)}{4}.$$

Substituting back:

$$\prod_{i=1}^n \|b_i\| \leq 2^{n(n-1)/4} \det(\mathcal{L}).$$

□

**Remark.** The ratio  $\frac{\prod_{i=1}^n \|b_i\|}{\det(\mathcal{L})}$  is called the orthogonality defect. For an LLL-reduced basis, this is bounded by a constant depending only on dimension.

**Proposition** (Approximation Factor). Let  $b_1, \dots, b_n$  be an LLL-reduced basis. Then for any non-zero  $v \in \mathcal{L}$ :

$$\|b_1\| \leq 2^{\frac{n-1}{2}} \|v\|.$$

Consequently, taking  $v$  such that  $\|v\| = \lambda_1(\mathcal{L})$ :

$$\|b_1\| \leq 2^{\frac{n-1}{2}} \lambda_1(\mathcal{L}).$$

*Proof.* Let  $v \in \mathcal{L} \setminus \{0\}$ . We can write  $v = \sum_{i=1}^n z_i b_i$  with  $z_i \in \mathbb{Z}$ . Let  $k$  be the largest index such that  $z_k \neq 0$ . Then  $v = \sum_{i=1}^k z_i b_i = z_k b_k^* + \sum_{i=1}^{k-1} (\dots) b_i^*$ . Since  $b_k^*$  is orthogonal to  $b_1^*, \dots, b_{k-1}^*$ , the squared norm is:

$$\|v\|^2 \geq z_k^2 \|b_k^*\|^2 \geq 1 \cdot \|b_k^*\|^2.$$

From the previous Lemma (setting  $i = 1, j = k$ ), we know  $\|b_1\|^2 \leq 2^{k-1} \|b_k^*\|^2$ . Substituting  $\|b_k^*\|^2 \leq \|v\|^2$ :

$$\|b_1\|^2 \leq 2^{k-1} \|v\|^2 \leq 2^{n-1} \|v\|^2.$$

Taking the square root gives  $\|b_1\| \leq 2^{(n-1)/2} \|v\|$ .

□

## 6 LLL Basis Reduction Algorithm

### The LLL Algorithm.

We present the basic version of the Lenstra-Lenstra-Lovász (LLL) algorithm. The algorithm works by iteratively enforcing the size-reduction condition (R1) and checking the Lovász condition (R2).

**Algorithm** (Basic LLL).

*Input:* Basis  $b_1, \dots, b_n \in \mathbb{Z}^n$  of  $\mathcal{L} \subseteq \mathbb{Z}^n$ .

*Output:* LLL-reduced basis of  $\mathcal{L}$ .

1. Compute GS orthogonalization  $b_1^*, \dots, b_n^*$  and coefficients  $\mu_{i,j}$  for  $1 \leq j < i \leq n$ .
2. For  $i = 2$  to  $n$ :
 

For  $j = i - 1$  down to 1:  $\triangleright$  size reduction

$$x := \lfloor \mu_{i,j} \rfloor$$

$$b_i := b_i - xb_j$$

$$\mu_{i,j} := \mu_{i,j} - x$$
 $\triangleright$  update coefficients
 For  $l = 1$  to  $j - 1$ :
 
$$\mu_{i,l} := \mu_{i,l} - x\mu_{j,l}$$

If  $\|b_i^*\|^2 < (\frac{3}{4} - \mu_{i,i-1}^2)\|b_{i-1}^*\|^2$ :  $\triangleright$  Lovász condition check

$$\text{Swap } b_i \text{ and } b_{i-1}.$$

$$\text{GOTO } 1.$$
3. Return  $b_1, \dots, b_n$ .

### Remarks.

- In the step 2., the operation  $b_i := b_i - xb_j$  is an elementary integer column operation. Thus, it preserves the lattice  $\mathcal{L}(b_1, \dots, b_n)$ .
- Since  $j < i$ , adding a multiple of  $b_j$  to  $b_i$  does not change the Gram-Schmidt vector  $b_i^*$ .
- The update ensures explicitly that the new  $|\mu_{i,j}| \leq 1/2$ .

### Stability of Orthogonalization (Lemmas).

We analyze how the Gram-Schmidt vectors and coefficients change during the Size Reduction step. Let  $B = (b_1, \dots, b_n)$  be a basis and  $C = (c_1, \dots, c_n)$  be a basis obtained by the operation:

$$c_j = b_j - xb_k \quad \text{where } k < j, \quad x \in \mathbb{R}$$

and  $c_i = b_i$  for all  $i \neq j$ .

**Lemma** (#1, Invariance of G-S Vectors). *Under the operation above,  $c_i^* = b_i^*$  for all  $1 \leq i \leq n$ .*

*Proof.* Recall that  $b_i^*$  is the projection of  $b_i$  onto the orthogonal complement of  $\text{span}(b_1, \dots, b_{i-1})$ . For  $i \neq j$ ,  $c_i = b_i$  and the span of previous vectors is identical, so  $c_i^* = b_i^*$ . For  $i = j$ ,  $c_j = b_j - xb_k$ . Since  $k < j$ ,  $b_k \in \text{span}(b_1, \dots, b_{j-1})$ . The projection is linear and maps any vector in the span to 0. Thus:

$$c_j^* = \text{proj}_{\langle b_1, \dots, b_{j-1} \rangle^\perp} (b_j - xb_k) = b_j^* - x \cdot 0 = b_j^*.$$

□

We now analyze the coefficients. Let  $\mu_{r,s} = \frac{\langle b_r, b_s^* \rangle}{\|b_s^*\|^2}$  denote coefficients for basis  $B$  and  $\delta_{r,s} = \frac{\langle c_r, c_s^* \rangle}{\|c_s^*\|^2}$  for basis  $C$ .

**Lemma** (#2,  $r \neq j$ ). *If  $r \neq j$ , then  $\delta_{r,s} = \mu_{r,s}$  for all valid  $s$ .*

*Proof.* If  $r \neq j$ , then  $c_r = b_r$ . By Lemma 1,  $c_s^* = b_s^*$ .

$$\delta_{r,s} = \frac{\langle c_r, c_s^* \rangle}{\|c_s^*\|^2} = \frac{\langle b_r, b_s^* \rangle}{\|b_s^*\|^2} = \mu_{r,s}.$$

□

**Lemma (#3,  $r = j, k < s < j$ ).** *If  $k < s < j$ , then  $\delta_{j,s} = \mu_{j,s}$ .*

*Proof.* Here  $c_j = b_j - xb_k$ .

$$\delta_{j,s} = \frac{\langle b_j - xb_k, b_s^* \rangle}{\|b_s^*\|^2} = \frac{\langle b_j, b_s^* \rangle}{\|b_s^*\|^2} - x \frac{\langle b_k, b_s^* \rangle}{\|b_s^*\|^2}.$$

Since  $k < s$ ,  $b_k \in \text{span}(b_1, \dots, b_{s-1})$ , so  $b_k$  is orthogonal to  $b_s^*$ . The second term is zero. Thus  $\delta_{j,s} = \mu_{j,s}$ . □

**Lemma (#4,  $r = j, s = k$ ).** *If  $s = k$ , then  $\delta_{j,k} = \mu_{j,k} - x$ .*

*Proof.* Using the expansion from the previous proof:

$$\delta_{j,k} = \mu_{j,k} - x \frac{\langle b_k, b_k^* \rangle}{\|b_k^*\|^2}.$$

Since  $b_k = b_k^* + \sum_{l < k} \mu_{k,l} b_l^*$ , and  $b_k^*$  is orthogonal to the sum,  $\langle b_k, b_k^* \rangle = \|b_k^*\|^2$ . Therefore,  $\delta_{j,k} = \mu_{j,k} - x \cdot 1$ . □

**Lemma (#5,  $r = j, s < k$ ).** *If  $s < k$ , then  $\delta_{j,s} = \mu_{j,s} - x\mu_{k,s}$ .*

*Proof.* Again,  $\delta_{j,s} = \mu_{j,s} - x \frac{\langle b_k, b_s^* \rangle}{\|b_s^*\|^2}$ . By definition,  $\frac{\langle b_k, b_s^* \rangle}{\|b_s^*\|^2} = \mu_{k,s}$ . Thus  $\delta_{j,s} = \mu_{j,s} - x\mu_{k,s}$ . □

**Remark.** These lemmas justify the update rules in Step 2 of the algorithm. Specifically, Lemma 4 justifies  $\mu_{i,j} := \mu_{i,j} - x$  (where  $j$  in the loop is  $k$  in the lemma), and Lemma 5 justifies the updates for  $l < j$ . Lemma 3 ensures we don't need to update coefficients for indices between  $j$  and  $i$ .

## Preparation for Termination Analysis.

To show the algorithm terminates, we introduce a potential function based on the GS norms.

**Definition.** For a basis  $b_1, \dots, b_n$ , define:

$$d_i = \prod_{j=1}^i \|b_j^*\|^2 \quad \text{for } i = 1, \dots, n, \quad \text{and } d_0 = 1.$$

The potential of the basis is defined as  $D = \prod_{i=1}^{n-1} d_i$ .

**Remark.** Note that  $d_n = \prod_{j=1}^n \|b_j^*\|^2 = \det(\mathcal{L})^2$ . Since  $\mathcal{L} \subseteq \mathbb{Z}^n$ ,  $d_n$  is a positive integer.

**Lemma.** For integer lattice bases  $b_1, \dots, b_n \in \mathbb{Z}^n$ , we have  $d_i \in \mathbb{N}$  for all  $i$ .

*Proof.* Recall the GS relation  $b_t = b_t^* + \sum_{s=1}^{t-1} \mu_{t,s} b_s^*$ . We can express the relation between the first  $i$  basis vectors and the first  $i$  Gram-Schmidt vectors as a matrix multiplication:

$$\underbrace{(b_1 | b_2 | \dots | b_i)}_{B_i \in M_{n,i}(\mathbb{Z})} = \underbrace{(b_1^* | b_2^* | \dots | b_i^*)}_{B_i^* \in M_{n,i}(\mathbb{R})} \begin{pmatrix} 1 & \mu_{2,1} & \cdots & \mu_{i,1} \\ 0 & 1 & \cdots & \mu_{i,2} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$



Let  $U_i \in M_{i,i}(\mathbb{R})$  be the upper triangular matrix defined above. The equation becomes:

$$B_i = B_i^* U_i.$$

Now, consider the Gram matrix of the basis vectors,  $G = B_i^T B_i$ . Since  $b_1, \dots, b_i \in \mathbb{Z}^n$ , the entries of  $G$  are scalar products of integer vectors, i.e., integers. Thus,  $\det(G) \in \mathbb{Z}$ . Substituting the factorization:

$$G = B_i^T B_i = (B_i^* U_i)^T (B_i^* U_i) = U_i^T (B_i^*)^T B_i^* U_i.$$

Taking the determinant:

$$\det(G) = \det(U_i^T) \det((B_i^*)^T B_i^*) \det(U_i).$$

Since  $U_i$  is upper triangular with 1s on the diagonal,  $\det(U_i) = \det(U_i^T) = 1$ . The matrix  $(B_i^*)^T B_i^*$  is diagonal because the columns of  $B_i^*$  are mutually orthogonal vectors. The diagonal entries are  $\langle b_j^*, b_j^* \rangle = \|b_j^*\|^2$ . Therefore:

$$\det((B_i^*)^T B_i^*) = \prod_{j=1}^i \|b_j^*\|^2 = d_i.$$

Combining these results, we get:

$$d_i = \det(G) \in \mathbb{Z}.$$

Since  $d_i$  is a product of squared norms of non-zero vectors,  $d_i > 0$ . Hence  $d_i \in \mathbb{N}$ . □

## 7 Complexity of LLL and Small Roots

### Number of Iterations.

We analyze the number of iterations of the LLL algorithm using the potential function  $D$ . Recall that  $d_i = \prod_{j=1}^i \|b_j^*\|^2$  and  $D = \prod_{i=1}^{n-1} d_i$ .

In the algorithm, the potential  $D$  changes only when a swap occurs between  $b_i$  and  $b_{i-1}$ . When  $b_i$  and  $b_{i-1}$  are swapped,  $d_{i-1}$  reduces by a factor of at least  $3/4$ , while other  $d_k$  remain unchanged. Consequently, the value of  $D$  decreases by a factor  $\leq 3/4$ .

**Proposition.** *The number of swap operations is polynomially bounded by the size of the input.*

*Proof.* Let  $D_0$  be the initial value of the potential. Let  $D_{fin}$  be the value at termination. Since  $D \geq 1$  (as  $d_i \in \mathbb{N}$ ), the number of iterations  $k$  satisfies:

$$1 \leq D_{fin} \leq \left(\frac{3}{4}\right)^k D_0 \implies k \leq \frac{\log D_0}{\log(4/3)}.$$

We estimate  $D_0$ . Let  $M' = \max_i \|b_i\|_{init}$ . Using  $\|b_j^*\|^2 \leq \|b_j\|^2 \leq (M')^2$ , we have  $d_i \leq (M')^{2i}$ .

$$D_0 = \prod_{i=1}^{n-1} d_i \leq \prod_{i=1}^{n-1} (M')^{2i} = (M')^{2 \sum_{i=1}^{n-1} i} = (M')^{n(n-1)}.$$

Let  $M = \max\{n, \log M'\}$  be the input size parameter. Then  $\log D_0 \leq n(n-1) \log M' \leq n^2 M$ . Thus, the number of iterations is  $O(n^2 M)$ , which is polynomial in the input size.  $\square$

**Remark.** *In each iteration, we perform arithmetic operations (scalar products, updates). The number of operations per iteration is roughly bounded by  $2\binom{n}{2}$  (updating coefficients).*

### Bit Complexity (Size of Numbers).

To prove the algorithm is polynomial time, we must ensure the numbers stored during the execution do not grow exponentially.

**Lemma.** *At any step of the algorithm  $d_{i-1}b_i^* \in \mathbb{Z}^n$ ,  $\|b_i^*\| \leq D$  and  $d_j\mu_{i,j} \in \mathbb{Z}$  for all  $1 \leq j < i \leq n$ .*

*Proof.* Recall that  $b_i^* = b_i - \sum_{j=1}^{i-1} a_j b_j$  where the coefficients  $a_1, \dots, a_{i-1} \in \mathbb{Q}$  are the solution of a system of linear equations:

$$G_{b_1, \dots, b_{i-1}}(a_1, \dots, a_{i-1})^T = (\langle b_i, b_1 \rangle, \dots, \langle b_i, b_{i-1} \rangle)^T.$$

Using Cramer's rule,  $a_j = \frac{\det(G_j)}{\det(G)}$ , where  $\det(G) = d_{i-1}$ . Thus,  $d_{i-1}a_j \in \mathbb{Z}$ , which implies

$$d_{i-1}b_i^* = d_{i-1}b_i - \sum (d_{i-1}a_j)b_j \in \mathbb{Z}^n.$$

Since  $d_{i-1}b_i^* \in \mathbb{Z}^n$ , if it is non-zero (which it is for a basis), its squared norm is at least 1:

$$\|d_{i-1}b_i^*\|^2 \geq 1 \implies d_{i-1}^2 \|b_i^*\|^2 \geq 1 \implies \|b_i^*\|^2 \geq \frac{1}{d_{i-1}^2}. \quad (*)$$

So we get that

$$\|b_i^*\|^2 = \frac{d_i}{d_{i-1}} = \frac{d_i}{\|b_1^*\|^2 \dots \|b_{i-1}^*\|^2} \stackrel{*}{\leq} d_i \cdot d_0^2 \dots d_{i-2}^2 \leq D^2 \implies \|b_i^*\| \leq D.$$

Finally, for the coefficients:

$$\mu_{i,j} = \frac{\langle b_i, b_j^* \rangle}{\|b_j^*\|^2} = \frac{\langle b_i, b_j^* \rangle}{\frac{d_j}{d_{j-1}}} = \frac{\langle b_i, d_{j-1}b_j^* \rangle}{d_j}.$$

Thus  $d_j\mu_{i,j} = \langle b_i, d_{j-1}b_j^* \rangle$ . Since  $b_i \in \mathbb{Z}^n$  and  $d_{j-1}b_j^* \in \mathbb{Z}^n$ , the product is an integer.  $\square$

**Lemma.** If vectors  $b_1, \dots, b_n$  satisfy condition (R1), i.e.,  $|\mu_{i,j}| \leq \frac{1}{2}$ , then  $\|b_i\|^2 \leq nD$ .

*Proof.*

$$\|b_i\|^2 = \|b_i^*\|^2 + \sum_{j=1}^{i-1} \mu_{i,j}^2 \|b_j^*\|^2 \leq \|b_i^*\|^2 + \frac{1}{4} \sum_{j=1}^{i-1} \|b_j^*\|^2 \leq D + \frac{n}{4} D \leq nD$$

Since  $\|b_k^*\|$  are bounded essentially by the input size (or values of  $d_k$ ),  $\|b_i\|$  remains polynomially bounded in bits.  $\square$

**Theorem.** The LLL algorithm runs in polytime with respect to the size of the input basis (dimension  $n$  and bit-length  $M$ ).

## Finding Small Roots of Polynomials.

We switch topics to solving polynomial equations modulo  $N$ , with applications to RSA.

**Problem:** Let  $N > 1$ . Let  $f(x) \in \mathbb{Z}[x]$  be a monic polynomial of degree  $d$ . Given a bound  $B > 0$ , we want to find all  $y \in \mathbb{Z}$  such that:

$$f(y) \equiv 0 \pmod{N} \quad \text{and} \quad |y| \leq B.$$

Usually, we are interested in the case where  $B \approx N^{1/d}$ .

*Note:* If we knew the factorization of  $N$ , we could solve this easily. We assume factorization is unknown.

**Definition.** For a polynomial  $g(x) = \sum_{i=0}^l g_i x^i$ , we define the coefficient vector:

$$v(g) = (g_0, g_1, \dots, g_l)^T \in \mathbb{Z}^{l+1}.$$

The norm  $\|g\|$  is defined as  $\|v(g)\|$ .

**Lemma** (Coppersmith / Howgrave-Graham). Let  $g(x) \in \mathbb{Z}[x]$  be a polynomial of degree  $\leq l$ . Let  $B, N \in \mathbb{N}$ . Assume:

$$\|v(g(Bx))\| < \frac{N}{\sqrt{l+1}}.$$

If there exists  $y \in \mathbb{Z}$  such that  $g(y) \equiv 0 \pmod{N}$  and  $|y| \leq B$ , then  $g(y) = 0$  (equality over  $\mathbb{Z}$ ).

*Proof.* We want to bound the absolute value of  $g(y)$ .

$$|g(y)| = \left| \sum_{i=0}^l g_i y^i \right| \leq \sum_{i=0}^l |g_i| |y|^i \leq \sum_{i=0}^l |g_i| B^i.$$

Notice that the coefficients of the polynomial  $g(Bx)$  are exactly  $g_i B^i$ . By Cauchy-Schwarz inequality applied to the vectors  $(|g_0|, |g_1|B, \dots, |g_l|B^l)$  and  $(1, 1, \dots, 1)$ :

$$\sum_{i=0}^l |g_i| B^i \leq \sqrt{\sum_{i=0}^l (g_i B^i)^2} \cdot \sqrt{\sum_{i=0}^l 1^2} = \|v(g(Bx))\| \cdot \sqrt{l+1}.$$

By the assumption  $\|v(g(Bx))\| < \frac{N}{\sqrt{l+1}}$ , we get:

$$|g(y)| < \frac{N}{\sqrt{l+1}} \cdot \sqrt{l+1} = N.$$

Since  $g(y) \equiv 0 \pmod{N}$ ,  $g(y)$  must be a multiple of  $N$ . The only multiple of  $N$  strictly between  $-N$  and  $N$  is 0. Thus,  $g(y) = 0$ .  $\square$

## 8 Small Integer Zeros of Polynomials & RSA Attacks

### RSA Cryptosystem Setup.

Let  $N = p \cdot q$  where  $p, q$  are large primes. We choose integers  $r, s$  such that:

$$r \cdot s \equiv 1 \pmod{\varphi(N)}$$

where  $\varphi(N) = (p-1)(q-1)$ .

- **Public Key:**  $(N, r)$  (Modulus, Public Exponent).
- **Private Key:**  $(N, s)$  (Modulus, Private Exponent).
- **Encryption:** Given a message  $M \in \mathbb{Z}_N$ , the ciphertext is  $C \equiv M^r \pmod{N}$ .
- **Decryption:**  $M \equiv C^s \pmod{N}$ .

### Håstad's Broadcast Attack (Motivation).

Assume a small public exponent  $r = 3$ . Scenario: Alice wants to send the same message  $M$  to three recipients: Bob, Charlie, and David.

- Bob's key:  $(N_B, 3)$ , ciphertext  $C_B \equiv M^3 \pmod{N_B}$ .
- Charlie's key:  $(N_C, 3)$ , ciphertext  $C_C \equiv M^3 \pmod{N_C}$ .
- David's key:  $(N_D, 3)$ , ciphertext  $C_D \equiv M^3 \pmod{N_D}$ .

Assume  $\gcd(N_B, N_C, N_D) = 1$ . By the Chinese Remainder Theorem (CRT), we can compute a value  $C'$  such that:

$$C' \equiv M^3 \pmod{N_B N_C N_D}.$$

Since  $M < N_i$ , we have  $M^3 < N_B N_C N_D$ . Therefore, the modular equality implies equality over integers:  $C' = M^3$ . We can simply compute the integer cube root of  $C'$  to recover  $M$ .

*This motivates the study of finding small roots of polynomials modulo  $N$ .*

### Coppersmith's Theorem.

**Theorem.** Let  $N \in \mathbb{Z}$  and  $f(x) \in \mathbb{Z}_N[x]$  be a monic polynomial of degree  $d$ . There exists an efficient algorithm to find all integers  $x_0$  such that:

$$f(x_0) \equiv 0 \pmod{N} \quad \text{and} \quad |x_0| \leq N^{1/d}.$$

We want to find the root  $x_0$  of  $f(x) \pmod{N}$ .

1. Construct a polynomial  $g(x) \in \mathbb{Z}[x]$  such that:

$$\{y \in \mathbb{Z} \mid f(y) \equiv 0 \pmod{N}, |y| \leq B\} \subseteq \{y \in \mathbb{Z} \mid g(y) = 0\}.$$

This condition is satisfied if  $g(x_0) \equiv 0 \pmod{N}$  and  $|g(x_0)| < N$ .

2. Use LLL reduction to find such a  $g(x)$  with small coefficients (implying small norm).
3. Compute roots of  $g(x)$  over integers (e.g., using Newton's method) and check if they are roots of  $f(x) \pmod{N}$ .

We analyze algorithms to find small integer roots of a monic polynomial  $f(x) \in \mathbb{Z}[x]$  of degree  $d$  modulo  $N$ . We seek a bound  $B$  such that we can efficiently find all roots  $|x_0| \leq B$ .

Simply said, we are gradually improving the design of the lattice in order to find larger and larger roots.

### Try 1: Linear Lattice.

Fix  $f(x) = x^d + a_{d-1}x^{d-1} + \dots + a_1x + a_0 \in \mathbb{Z}[x]$  and consider the set of polynomials

$$\mathcal{Z}_1 = \{\underbrace{N}_{g_0}, \underbrace{Nx}_{g_1}, \dots, \underbrace{Nx^{d-1}}_{g_{d-1}}, \underbrace{f(x)}_{g_d}\}.$$

We consider integer combinations of  $\mathcal{Z}_1$  to form a new polynomial  $g = \sum_{g_i \in \mathcal{Z}_1} y_i g_i$ , where  $y_i \in \mathbb{Z}$ . Note that if  $f(y) \equiv 0 \pmod{N}$  for some  $y$ , then  $g(y) \equiv 0 \pmod{N}$ , because  $g(x) \equiv y_d f(x) \pmod{N}$  (other terms are multiples of  $N$ ). This equivalence holds if  $\gcd(y_d, N) = 1$ .

We consider a lattice  $\mathcal{L}$  with basis vectors  $v(g_0(Bx)), v(g_1(Bx)), \dots, v(g_d(Bx))$ . The basis matrix  $M$  with the dimension  $d+1$  (where columns represent the basis vectors) is:

$$M = \begin{pmatrix} N & 0 & 0 & \dots & a_0 \\ 0 & NB & 0 & \dots & a_1 B \\ \vdots & \vdots & \ddots & & \vdots \\ 0 & 0 & \dots & NB^{d-1} & a_{d-1} B^{d-1} \\ 0 & 0 & \dots & 0 & B^d \end{pmatrix}$$

The determinant of the lattice is the determinant of this upper-triangular matrix:

$$\det(\mathcal{L}) = N^d \cdot B^{0+1+\dots+d} = N^d B^{\frac{d(d+1)}{2}}.$$

We apply the LLL algorithm to compute a reduced basis  $b_1, \dots, b_{d+1}$ . The first vector satisfies:

$$\|b_1\| \leq 2^{\frac{\dim-1}{4}} \det(\mathcal{L})^{\frac{1}{\dim}} = 2^{\frac{d}{4}} (N^d B^{\frac{d(d+1)}{2}})^{\frac{1}{d+1}} = 2^{d/4} N^{\frac{d}{d+1}} B^{d/2}.$$

We need to figure out for which value of  $B$  we have  $\|v(g(Bx))\| < \frac{N}{\sqrt{d+1}}$ . Using the LLL bound:

$$\begin{aligned} 2^{d/4} N^{\frac{d}{d+1}} B^{d/2} < \frac{N}{\sqrt{d+1}} &\iff B^{d/2} < \frac{N^{1-\frac{d}{d+1}}}{2^{d/4} \sqrt{d+1}} = \frac{N^{\frac{1}{d+1}}}{2^{d/4} \sqrt{d+1}}. \\ &\stackrel{(*)}{\iff} B < \frac{N^{\frac{2}{d(d+1)}}}{\sqrt{2}(d+1)^{1/d}} \iff B < c_d N^{\frac{2}{d(d+1)}} \iff B \approx N^{2/d^2} \end{aligned}$$

### Try 2: Enlarged Lattice.

Consider the larger set

$$\mathcal{Z}_2 = \{N, Nx, \dots, Nx^{d-1}\} \cup \{f(x), xf(x), \dots, x^{d-1}f(x)\}.$$

If  $y \in \mathbb{Z}$  satisfies  $f(y) \equiv 0 \pmod{N}$ , then  $g(y) \equiv 0 \pmod{N}$  where  $g = \sum y_i g_i$ , since all  $g_i$  are multiples of  $N$  or  $f(x)$ .

Let  $\mathcal{L} \subseteq \mathbb{Z}^{2d}$  be a lattice with basis  $v(g_0(Bx)), \dots, v(g_{2d-1}(Bx))$ . There is a corresponding matrix of dimension  $2d \times 2d$ . The diagonal entries are:

$$M = \begin{pmatrix} N & 0 & \dots & 0 & 0 & 0 & 0 & 0 \\ 0 & NB & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & 0 & \ddots & \ddots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & NB^{d-1} & 0 & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & 0 & B^d & 0 & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & 0 & B^{d+1} & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \dots & 0 & 0 & 0 & 0 & B^{2d-1} \end{pmatrix}$$

The determinant is:

$$\det(\mathcal{L}) = N^d \cdot B^{0+1+\dots+(2d-1)} = N^d B^{\frac{2d(2d-1)}{2}} = N^d B^{d(2d-1)}.$$

We apply LLL to find a reduced basis  $b_1, \dots, b_{2d}$ . If we find a vector  $b_1$  corresponding to  $g(x)$ , we have the bound:

$$\|b_1\| \leq 2^{\frac{2d-1}{4}} \det(\mathcal{L})^{\frac{1}{2d}} = 2^{\frac{2d-1}{4}} (N^d B^{d(2d-1)})^{\frac{1}{2d}} = 2^{\frac{2d-1}{4}} N^{1/2} B^{\frac{2d-1}{2}}.$$

We want to figure out the correct size for  $B$  such that we can apply the *Howgrave-Graham lemma* from previous lecture (we need  $\|b_1\| < \frac{N}{\sqrt{2d}}$ ):

$$\begin{aligned} 2^{\frac{2d-1}{4}} N^{1/2} B^{\frac{2d-1}{2}} &< \frac{N}{\sqrt{2d}} \iff B^{\frac{2d-1}{2}} < \frac{N^{1/2}}{2^{\frac{2d-1}{4}} \sqrt{2d}} \\ &\iff B < \frac{N^{\frac{1}{2d-1}}}{\sqrt{2}(2d)^{1/(2d-1)}} \iff B < c_d \cdot N^{\frac{1}{2d-1}}. \end{aligned}$$

This allows finding roots up to  $B \approx N^{1/(2d-1)}$ .

### Try 3: Coppersmith's Method (Modulo $N^m$ ).

Fix  $B, m \in \mathbb{N}$  and consider the set

$$\mathcal{Z}_3 = \{N^{m-i-1} f(x)^i x^j \mid 0 \leq j < d, 0 \leq i < m\}.$$

In case  $m = 2$ , we get  $\mathcal{Z}_3 = \mathcal{Z}_2$  and we are done. Note that  $\mathcal{Z}_3$  consists of the following polynomials:

- $N^{m-1}, N^{m-1}x, \dots, N^{m-1}x^{d-1}$  (for  $i = 0$ )
- $N^{m-2}f(x), N^{m-2}f(x)x, \dots, N^{m-2}f(x)x^{d-1}$  (for  $i = 1$ )
- ...
- $f(x)^{m-1}, f(x)^{m-1}x, \dots, f(x)^{m-1}x^{d-1}$  (for  $i = m-1$ )

We denote all polynomials in the set by  $g_0, g_1, \dots, g_{md-1}$ . Given  $g = \sum y_k g_k$  with  $y_k \in \mathbb{Z}$ . If  $f(y) \equiv 0 \pmod{N}$  for some  $y \in \mathbb{Z}$ , then  $f(y) = kN$  for some integer  $k$ . Consequently:

$$g_k(y) = N^{m-i-1} (kN)^i y^j = k^i N^{m-1} y^j \equiv 0 \pmod{N^{m-1}}.$$

Since  $g$  is a linear combination of these,  $g(y) \equiv 0 \pmod{N^{m-1}}$ .

The dimension of the lattice is  $w = md$ . The determinant is the product of the leading terms evaluated at  $Bx$ :

$$\det(\mathcal{L}) = \prod_{i=0}^{m-1} \prod_{j=0}^{d-1} N^{m-i-1} B^{di+j} = N^{\frac{dm(m-1)}{2}} B^{\frac{dm(dm-1)}{2}}.$$

The LLL condition for the shortest vector  $b_1$  gives  $\|b_1\| \approx \det(\mathcal{L})^{1/w}$ . We need  $\|b_1\| < \frac{N^{m-1}}{\sqrt{w}}$  to imply  $g(y) = 0$  over integers. Approximating and ignoring constants:

$$N^{\frac{m-1}{2}} B^{\frac{dm-1}{2}} < N^{m-1}.$$

$$B^{\frac{dm-1}{2}} < N^{\frac{m-1}{2}} \implies B < N^{\frac{m-1}{dm-1}}.$$

For large  $m$ , the exponent approaches  $\frac{1}{d}$ . Thus, we can find roots up to  $B \approx N^{1/d}$ .

## 9 CVP and Nearest Plane Algorithm

### Håstad's Broadcast Attack (Generalization).

We revisit the RSA broadcast attack scenario where a message  $M$  is encrypted and sent to  $r$  different recipients with public keys  $(N_1, e), \dots, (N_r, e)$  where  $e = d$  is the degree of the polynomial.

**Theorem.** Let  $N_1, \dots, N_r$  be pairwise coprime integers and let  $N_{\min} = \min\{N_1, \dots, N_r\}$ . Let  $g_i \in \mathbb{Z}_{N_i}[x]$  be  $r$  polynomials of degree  $d_i \leq d$ . Suppose there exists a unique  $M < N_{\min}$  such that

$$g_i(M) \equiv C_i \pmod{N_i} \quad \text{for } i = 1, \dots, r.$$

Then if  $r \geq d$ , one can efficiently find  $M$  given  $\{(N_i, g_i, C_i)\}_{i=1}^r$ .

*Proof.* For each  $i \in \{1, \dots, r\}$ , define  $h_i(x) = g_i(x) - C_i$ . We are looking for  $M$  such that  $h_i(M) \equiv 0 \pmod{N_i}$ . Without loss of generality, assume  $h_i$  are monic of degree  $d$ .

We apply the Chinese Remainder Theorem (CRT) to coefficients to construct a polynomial  $h(x)$  modulo  $N = \prod_{i=1}^r N_i$ . The CRT construction gives:

$$h(x) = \sum_{i=1}^r T_i h_i(x) \pmod{N}$$

where  $T_i \equiv 1 \pmod{N_i}$  and  $T_i \equiv 0 \pmod{N_j}$  for  $j \neq i$ . Note that  $h(x)$  has degree  $d$  and is monic because the coefficient at  $x^d$  is  $\sum T_i \cdot 1 \equiv 1 \pmod{N_i}$  for all  $i$ , hence  $\equiv 1 \pmod{N}$ .

Finally, since  $h_i(M) \equiv 0 \pmod{N_i}$  for all  $i$ , we have  $h(M) \equiv 0 \pmod{N}$ . Using the *root-finding theorem* (Coppersmith's method), there exists an efficient algorithm that finds the set

$$\{y \in \mathbb{Z} \mid h(y) \equiv 0 \pmod{N}, |y| < N^{1/d}\}.$$

We need to verify that our  $M$  falls into this range. Since  $r \geq d$ , we have:

$$M < N_{\min} \leq N^{1/r} \leq N^{1/d}.$$

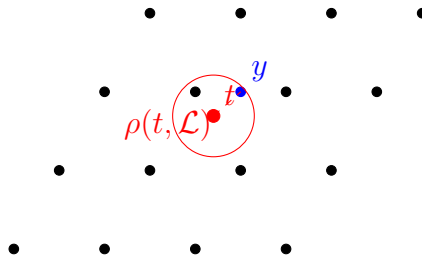
Thus, the algorithm efficiently recovers  $M$ . □

### Closest Vector Problem (CVP).

**Definition** (Closest vector). Let  $\mathcal{L} \subset \mathbb{R}^n$  be a lattice. For a target vector  $t \in \mathbb{R}^n$ , we define the distance from  $t$  to the lattice as:

$$\rho(t, \mathcal{L}) = \min\{\|t - v\| : v \in \mathcal{L}\}.$$

A vector  $y \in \mathcal{L}$  is called a closest vector to  $t$  if  $\|y - t\| = \rho(t, \mathcal{L})$ .



For  $\mathcal{L} \subseteq \mathbb{Z}^n, t \in \mathbb{Z}^n$  and fixed  $\gamma \geq 1$  we define the following computational problems:

- **CVP $_{\gamma}$  (Search):** Find  $v \in \mathcal{L}$  such that  $\|v - t\| \leq \gamma \cdot \rho(t, \mathcal{L})$ .
- **CVP $_{\gamma}$  (Optimization):** Find  $d$  such that  $\rho(t, \mathcal{L}) \leq d \leq \gamma \cdot \rho(t, \mathcal{L})$ .
- **CVP $_{\gamma}$  (Decision):** Given  $r$ , decide if  $\rho(t, \mathcal{L}) \leq r$  or  $\rho(t, \mathcal{L}) > \gamma r$ .

## Nearest Plane Algorithm.

**Motivation:** Given a lattice  $\mathcal{L}$  and basis  $b_1, \dots, b_n$ , we can decompose the lattice into a union of "layers" (cosets of the sublattice  $\mathcal{L}' = \mathcal{L}(b_1, \dots, b_{n-1})$ ).

$$\mathcal{L} = \bigcup_{c \in \mathbb{Z}} (\mathcal{L}' + cb_n).$$

These layers are hyperplanes separated by distance  $\|b_n^*\|$ . The algorithm greedily selects the hyperplane closest to  $t$ , projects  $t$  onto it, and recurses.

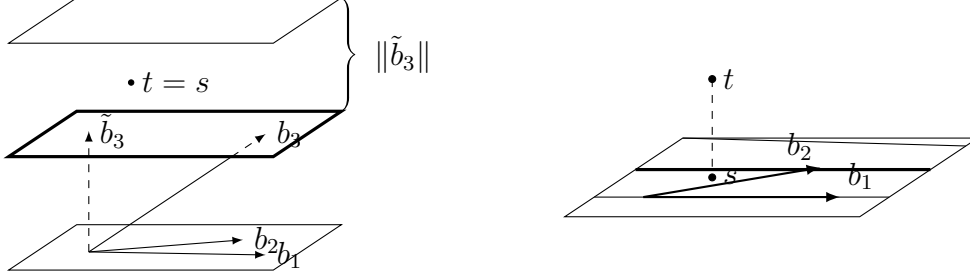


Figure 1: *The nearest plane algorithm for a rank 3 lattice and the resulting rank 2 instance. The chosen hyperplanes are thicker.*

**Algorithm** (Nearest Plane Algorithm (NPA)).

*Input:* LLL-reduced basis  $B = (b_1, \dots, b_n)$ , target  $t \in \mathbb{R}^n$ .

*Output:* Vector  $x \in \mathcal{L}$  approximating the closest vector.

1. Find  $b_1, \dots, b_n$  LLL-reduced basis of  $\mathcal{L}$ .
2.  $b \leftarrow t$
3. For  $j = n$  down to 1:  
 $b \leftarrow b - c_j b_j$ , where  $c_j = \lfloor \frac{\langle b, b_j^* \rangle}{\|b_j^*\|^2} \rfloor$ .
4. Return  $t - b$ .

**Analysis.** Let  $\langle \mathcal{L} \rangle_{\mathbb{R}} = \text{span}(b_1, \dots, b_n)$ . It suffices to work with the orthogonal projection of  $t$  onto the span of the lattice. Let  $b_i$  be LLL-reduced and  $b_i^*$  be the GSO vectors. Any vector  $u \in \text{span}(\mathcal{L})$  can be written in the orthogonal basis:  $u = \sum x_i b_i^*$  where  $x_i = \frac{\langle u, b_i^* \rangle}{\|b_i^*\|^2}$ .

**Proposition** (Coefficient Bounds). *The values  $c_1, \dots, c_n$  chosen by the recursive calls satisfy:*

$$t - \sum_{i=1}^n c_i b_i = \sum_{i=1}^n x_i b_i^*$$

where  $|x_i| \leq \frac{1}{2}$ .

Basically, the algorithm rounds the coefficient of  $b_n^*$  at each step, making the "error" in that direction at most  $1/2$ .

**Lemma** (Claim 1). *For any  $t \in \langle \mathcal{L} \rangle_{\mathbb{R}}$ , the output  $x$  of NPA satisfies:*

$$\|x - t\|^2 \leq \frac{1}{4} \sum_{i=1}^n \|b_i^*\|^2.$$

*Proof.* Let  $e = x - t$ . From the proposition above,  $e = \sum_{i=1}^n x_i b_i^*$  with  $|x_i| \leq 1/2$ . Since  $b_i^*$  are pairwise orthogonal:

$$\|e\|^2 = \left\| \sum_{i=1}^n x_i b_i^* \right\|^2 = \sum_{i=1}^n x_i^2 \|b_i^*\|^2 \leq \sum_{i=1}^n \frac{1}{4} \|b_i^*\|^2.$$

□



**Lemma** (Claim 2). *For any  $t \in \langle \mathcal{L} \rangle_{\mathbb{R}}$  and output  $x$  of NPA:*

$$\|x - t\| \leq \frac{1}{2} \cdot 2^{n/2} \|b_n^*\|.$$

*Proof.* We use Claim 1 and the property of LLL-reduced bases:  $\|b_i^*\|^2 \leq 2^{j-i} \|b_j^*\|^2$ . Specifically,  $\|b_i^*\|^2 \leq 2^{n-i} \|b_n^*\|^2$ . Summing the geometric series:

$$\|x - t\|^2 \leq \frac{1}{4} \sum_{i=1}^n 2^{n-i} \|b_n^*\|^2 = \frac{\|b_n^*\|^2}{4} \sum_{k=0}^{n-1} 2^k < \frac{\|b_n^*\|^2}{4} \cdot 2^n.$$

Taking the square root:

$$\|x - t\| < \frac{1}{2} \sqrt{2^n} \|b_n^*\| = \frac{1}{2} \cdot 2^{n/2} \cdot \|b_n^*\|.$$

□

**Theorem** (Correctness). *The NPA gives a  $2^{n/2}$ -approximation for CVP. That is, for any  $t \in \mathbb{R}^n$ , if  $y \in \mathcal{L}$  is the closest point to  $t$ , then the output  $x$  satisfies:*

$$\|x - t\| \leq 2^{n/2} \|y - t\|.$$

*Proof.* We prove this by induction on  $n$  (the rank). Let  $s$  be the orthogonal projection of  $t$  onto  $\text{span}(\mathcal{L})$ . Since  $\|x - t\|^2 = \|x - s\|^2 + \|s - t\|^2$  and similarly for  $y$ , it suffices to show  $\|x - s\| \leq 2^{n/2} \|y - s\|$ . Assume  $t \in \text{span}(\mathcal{L})$ .

- (i)  $n = 1$ : The algorithm rounds to the nearest integer, which is optimal.  $\|x - t\| \leq \|y - t\|$ .
- (ii)  $n > 1$ : Let  $H_c = \mathcal{L}(b_1, \dots, b_{n-1}) + cb_n$  be the hyperplanes. The algorithm selects the plane  $H_{c_n}$  closest to  $t$ . Let  $t'$  be the projection of  $t$  onto  $H_{c_n}$ . The closest vector  $y$  belongs to some  $H_k$ . There are two cases:

- **Case 1:**  $y \in H_{c_n}$ . Then  $y$  and  $x$  are both in the hyperplane chosen by the algorithm. By the induction hypothesis (applied to the  $(n-1)$ -dim lattice in  $H_{c_n}$ ):

$$\|x - t'\| \leq 2^{(n-1)/2} \|y - t'\|.$$

Since  $t - t'$  is orthogonal to  $H_{c_n}$ :

$$\begin{aligned} \|x - t\|^2 &= \|x - t'\|^2 + \|t' - t\|^2 \\ &\leq 2^{n-1} \|y - t'\|^2 + \|t' - t\|^2 \\ &\leq 2^{n-1} (\|y - t'\|^2 + \|t' - t\|^2) = 2^{n-1} \|y - t\|^2. \end{aligned}$$

Thus  $\|x - t\| \leq 2^{(n-1)/2} \|y - t\| \leq 2^{n/2} \|y - t\|$ .

- **Case 2:**  $y \notin H_{c_n}$ . The distance between adjacent hyperplanes is  $\|b_n^*\|$ . Since the algorithm picked the closest plane  $H_{c_n}$ , and  $y$  is in a different plane  $H_k$  ( $k \neq c_n$ ), the distance from  $t$  to  $y$  must be at least half the distance between planes:

$$\|y - t\| \geq \frac{1}{2} \|b_n^*\|.$$

On the other hand, from Claim 2, we know the algorithm outputs  $x$  such that:

$$\|x - t\| \leq \frac{1}{2} 2^{n/2} \|b_n^*\|.$$

Combining these:

$$\|x - t\| \leq 2^{n/2} \left( \frac{1}{2} \|b_n^*\| \right) \leq 2^{n/2} \|y - t\|.$$

□

## 10 Basic Complexity Results

### Search CVP to Decision CVP Reduction.

We show that if we have an oracle for the CVP(Decision) problem, we can solve the CVP(Search) (finding the actual closest vector) in polynomial time.

**Lemma.** *CVP(Search) can be solved in polynomial time given an oracle that solves CVP(Decision).*

*Proof.* Let  $\mathcal{L}$  be a lattice with basis  $B = \{b_1, \dots, b_n\}$  and  $t$  be a target vector.

- *Step 1: Determine the distance  $r$  (Optimization).* First, we find the exact distance  $r = \rho(t, \mathcal{L})$ . Let  $R = \sum_{i=1}^n \|b_i\|$ . This is a rough upper bound for the distance, because we can write  $t = \sum t_i b_i$  and approximate it by the lattice vector  $v = \sum \lfloor t_i \rfloor b_i$ . The distance  $\|t - v\|$  is bounded by  $R$ .

Using the CVP(Decision) oracle, we perform a binary search to find  $r$  such that  $\rho(t, \mathcal{L}) \leq r$  is YES and  $\rho(t, \mathcal{L}) < r$  is NO. This takes  $O(\log R)$  oracle calls.

- *Step 2: Find the vector (Iterative Step).* Once we have  $r$ , we want to find  $v \in \mathcal{L}$  such that  $\|t - v\| = r$ . The core idea is to determine the coefficients of the closest vector  $v = \sum x_i b_i$  bit by bit. We modify the basis to make the lattice “sparser” while keeping the target relative to the original lattice.

We iterate  $k$  times. In each step, we process the basis vectors. Let the current state be  $(B, t)$ . For each basis vector  $b_i$  in  $B$ :

1. Construct a sublattice basis  $B''$  by doubling the  $i$ -th vector:

$$B'' = \{b_1, \dots, b_{i-1}, 2b_i, b_{i+1}, \dots, b_n\}.$$

Note that  $\mathcal{L}(B'')$  contains only vectors where the  $i$ -th coefficient is even. The original lattice  $\mathcal{L}(B)$  is the union of two cosets:

$$\mathcal{L}(B) = \mathcal{L}(B'') \cup (\mathcal{L}(B'') + b_i).$$

2. Query the Oracle: Is  $\rho(t, \mathcal{L}(B'')) \leq r$ ?

3. Branching:

- **YES:** The closest vector lies in the sublattice  $\mathcal{L}(B'')$ . This means the coefficient of  $b_i$  for the closest vector is even. We update  $B \leftarrow B''$  (replace  $b_i$  with  $2b_i$ ). The target  $t$  remains unchanged.
- **NO:** The closest vector must be in the coset  $\mathcal{L}(B'') + b_i$ . This means the closest vector to  $t$  has the form  $v = v' + b_i$  where  $v' \in \mathcal{L}(B'')$ . The distance condition is  $\|t - (v' + b_i)\| \leq r$ , which implies  $\|(t - b_i) - v'\| \leq r$ . So,  $\rho(t - b_i, \mathcal{L}(B'')) \leq r$ . We update  $B \leftarrow B''$  (replace  $b_i$  with  $2b_i$ ) and shift the target  $t \leftarrow t - b_i$ .

- *Step 3: Termination.* We repeat this doubling process  $k$  times for every basis vector. At the end of this process, we obtain a basis  $B^* = \{2^k b_1, \dots, 2^k b_n\}$  and a modified target  $t^*$ . The invariants maintained are:

$$(1) \rho(t^*, \mathcal{L}(B^*)) = r.$$

$$(2) t^* = t - v_{shift}, \text{ where } v_{shift} \in \mathcal{L}(B) \text{ is the accumulation of all } b_i \text{ subtractions.}$$

We can use the NPA on  $(B^*, t^*)$ . Since the basis vectors are scaled by  $2^k$ , the lattice is extremely sparse and orthogonal-like relative to the small radius  $r$ . The NPA will return the lattice vector  $\mathbf{0}$  correctly because the approximation factor  $2^{n/2}$  is overwhelmed by the scaling factor  $2^k$ .

Thus, the closest vector in the sparse lattice is  $\mathbf{0}$ . The relation  $t^* \approx \mathbf{0}$  implies  $t - v_{shift} \approx \mathbf{0}$ , so  $v_{shift} \approx t$ . The closest vector in the original lattice is exactly  $v_{shift}$  (plus the result from the final step, which is  $\mathbf{0}$ ).

□

## NP-Completeness of CVP.

**Theorem.** *CVP(Decision) is NP-complete.*

We know CVP is in NP (certificate is the vector). We prove hardness by reducing from the SSP.

### Subset Sum Problem (SSP).

*Input:* Integers  $a_1, \dots, a_n \in \mathbb{Z}^+$  and a target sum  $S$ .

*Question:* Do there exist  $\epsilon_1, \dots, \epsilon_n \in \{0, 1\}$  such that  $\sum_{i=1}^n \epsilon_i a_i = S$ ?

**Reduction.** Given an instance of SSP, we construct a CVP instance  $(\mathcal{L}, t, r)$ . We define the lattice  $\mathcal{L} \subset \mathbb{Z}^{n+1}$  using the basis vectors (as columns of matrix  $B$ ):

$$B = \begin{pmatrix} a_1 & \dots & a_n \\ 2 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 2 \end{pmatrix}$$

The target vector is  $t = (S, 1, 1, \dots, 1)^T \in \mathbb{Z}^{n+1}$ . We set the decision radius to be  $r = \sqrt{n}$ .

**Claim.** *There exist  $\epsilon_i \in \{0, 1\}$  such that  $\sum \epsilon_i a_i = S \iff \rho(t, \mathcal{L}) \leq \sqrt{n}$ .*

*Proof.* We prove both implications.

“ $\Rightarrow$ ”  $S = \sum \epsilon_i a_i$  and  $r = \sum \epsilon_i b_i$ . Then  $r - t = (0, \pm 1, \dots, \pm 1) \implies \|r - t\| = \sqrt{n}$ .

“ $\Leftarrow$ ” Assume  $S \neq \sum \epsilon_i a_i$  for all choices of  $\epsilon \in \{0, 1\}$ . We take  $\mathcal{L} \ni r \in \sum y_i b_i$ , where  $y_i \in \mathbb{Z}$ .

So  $r - t = (\sum y_i a_i - S, \ell_1, \dots, \ell_n)$ , where  $\ell_i \in \mathbb{Z} \setminus \{0\}$ ,  $\sum_{i=1}^n \ell_i^2 \geq n$ .

The equality holds  $\iff \ell_i = \pm 1 \iff y_i \in \{0, 1\}$ .

If  $\sum \ell_i^2 = n$  and given that  $S - \sum \epsilon_i a_i \neq 0$ , then  $\|r - t\| > \sqrt{n}$ .

□

# 11 SVP vs CVP and The Sieve Algorithm

## Complexity Relations: SVP vs CVP.

We investigate the relative hardness of SVP and CVP. The aim is to show that solving SVP is “no harder” than solving CVP.

For an approximation factor  $\gamma \geq 1$  and given lattice  $\mathcal{L}$ , target  $t$ , and radius  $r$ :

- **GapCVP $_\gamma$** : Output YES if  $\rho(t, \mathcal{L}) \leq r$ , NO if  $\rho(t, \mathcal{L}) > \gamma r$ .
- **GapSVP $_\gamma$** : Output YES if  $\lambda_1(\mathcal{L}) \leq r$ , NO if  $\lambda_1(\mathcal{L}) > \gamma r$ .

**Remark.** If  $\rho(t, \mathcal{L}) \in (r, \gamma r]$ , the algorithm can return any answer (or error).

**Remark.** For  $\gamma = 1$ , GapCVP $_1$  is equivalent to the standard Decision CVP.

**Theorem.** For any  $\gamma \geq 1$ , given GapCVP $_\gamma$  oracle, it is impossible to solve GapSVP $_\gamma$  in polytime.

*Proof.* Let  $(\mathcal{L}, r)$  be an instance of GapSVP $_\gamma$ , where  $\mathcal{L}$  is generated by basis  $B = \{b_1, \dots, b_n\} \subset \mathbb{Z}^n$ . We construct  $n$  instances of GapCVP $_\gamma$ . For each  $i \in \{1, \dots, n\}$ , consider the sublattice  $\mathcal{L}_i$  generated by the basis:

$$B_i = \{b_1, \dots, b_{i-1}, 2b_i, b_{i+1}, \dots, b_n\}.$$

This sublattice contains all vectors  $v = \sum c_j b_j \in \mathcal{L}$  where the coefficient  $c_i$  is even. We set the target vector  $t_i = b_i$  and the distance  $r$ . We query the GapCVP $_\gamma$  oracle on inputs  $(\mathcal{L}_i, t_i, r)$  for all  $i$ .

- **YES case:** If  $\lambda_1(\mathcal{L}) \leq r$ , there exists a non-zero vector  $v = \sum c_j b_j$  with  $\|v\| \leq r$ . Since  $v \neq 0$ , at least one coefficient  $c_k$  is odd. We can write  $v = w + b_k$  (or  $w - b_k$ ) where  $w \in \mathcal{L}_k$  (since subtracting  $b_k$  makes the  $k$ -th coefficient even). Thus,  $\|v\| = \|w + b_k\| = \|(-w) - t_k\|$ . Since  $-w \in \mathcal{L}_k$ , we have  $\rho(t_k, \mathcal{L}_k) \leq r$ . The oracle for index  $k$  will return YES.
- **NO case:** If  $\lambda_1(\mathcal{L}) > \gamma r$ , then for any non-zero vector  $v \in \mathcal{L}$ ,  $\|v\| > \gamma r$ . Any vector  $u \in \mathcal{L}_i$  implies  $u - t_i = u - b_i \in \mathcal{L}$ . Moreover,  $b_i \notin \mathcal{L}_i$ , so  $u - b_i \neq 0$ . Therefore, the distance  $\|u - t_i\|$  corresponds to the length of some non-zero vector in  $\mathcal{L}$ . Hence  $\rho(t_i, \mathcal{L}_i) \geq \lambda_1(\mathcal{L}) > \gamma r$ . All oracles return NO.

□

## $2^{O(n)}$ -time Algorithm for SVP (AKS Sieve).

While LLL gives a  $2^n$ -approximation in polynomial time, we can solve SVP exactly (or with  $(1 + \epsilon)$  approximation) in time exponential in  $n$  but polynomial in the bit-size of entries. This is the Ajtai-Kumar-Sivakumar (AKS) algorithm.

**Lemma (Scaling).** Given an algorithm  $\mathcal{A}$  that finds the shortest non-zero vector in lattices satisfying  $2 \leq \lambda_1(\mathcal{L}) \leq 3$ , we can find the shortest non-zero vector in any lattice. The running time increases by at most a factor of  $O(n)$ .

*Proof.* Let  $\mathcal{L}$  be a lattice with basis  $B$ . We apply LLL algorithm to obtain an estimate  $\hat{\lambda}_1$  such that  $\lambda_1(\mathcal{L}) \leq \hat{\lambda}_1 \leq 2^n \lambda_1(\mathcal{L})$ . We construct a sequence of scaled bases  $B_i$  for  $i = 0, \dots, 2n$ :

$$B_i = \frac{1.5^i}{\hat{\lambda}_1} B.$$

Let  $\mathcal{L}_i$  be the lattice generated by  $B_i$ . Then  $\lambda_1(\mathcal{L}_i) = \frac{1.5^i}{\hat{\lambda}_1} \lambda_1(\mathcal{L})$ . Since  $1 \leq \frac{\hat{\lambda}_1}{\lambda_1(\mathcal{L})} \leq 2^n$ , there exists some  $i$  such that the scaled shortest vector length falls into the interval  $[2, 3]$ . We run algorithm  $\mathcal{A}$  on all  $\mathcal{L}_i$ . For the correct  $i$ , it returns a vector  $v_i$ . We can recover the original vector by rescaling:  $v = \frac{\hat{\lambda}_1}{1.5^i} v_i$ . □

**Lemma (Sieve).** *Let  $R > 0$ ,  $I = \{1, \dots, N\}$ . For any set of points  $X = \{x_1, \dots, x_N\} \subset B(0, R)$ , we can find a subset  $J \subseteq I$  and a map  $\varphi : I \rightarrow J$  such that:*

$$(1) |J| \leq 5^n.$$

$$(2) \forall i \in I : \|x_i - x_{\varphi(i)}\| \leq \frac{R}{2}.$$

*The running time is polynomial in  $N$  and dimensions.*

*Proof.* We construct the following:

1. Initialize  $J = \emptyset$ .
2. For  $i = 1$  to  $N$ :  
 If there exists  $j \in J$  such that  $\|x_i - x_j\| \leq R/2$ , set  $\varphi(i) = j$ .  
 Else, set  $J = J \cup \{i\}$  and  $\varphi(i) = i$ .
3. Return  $J, \varphi$

By construction, for any distinct  $j, j' \in J$ ,  $\|x_j - x_{j'}\| > R/2$ . Consider the open balls of radius  $R/4$  centered at points in  $J$ . These balls are disjoint. Also, since  $x_j \in B(0, R)$ , these small balls are contained in  $B(0, R + R/4) = B(0, \frac{5}{4}R)$ . Comparing volumes:

$$\begin{aligned} |J| \cdot \text{vol}(B(0, \frac{R}{4})) &\leq \text{vol}(B(0, \frac{5}{4}R)) \\ |J| \cdot (\frac{R}{4})^n \text{vol}(B(0, 1)) &\leq (\frac{5}{4}R)^n \text{vol}(B(0, 1)) \\ |J| &\leq \frac{(5/4)^n}{(1/4)^n} = 5^n. \end{aligned}$$

□

## The Sieve Algorithm.

**Algorithm** (Sieve for SVP).

*Input:* Lattice  $\mathcal{L} \subseteq \mathbb{Z}^n$  with basis  $B = \{b_1, \dots, b_n\}$ , such that  $2 \leq \lambda_1(\mathcal{L}) \leq 3$ .

*Output:* A shortest vector  $v \in \mathcal{L}$ , such that  $\|v\| = \lambda_1(\mathcal{L})$ .

1.  $R_0 \leftarrow n \max_i \|b_i\|$   
 Choose  $N = 2^{8n} \log R_0$  points  $x_1, \dots, x_N$  uniformly in  $B(0, 2)$   
 Compute  $y_i \equiv x_i \bmod \Phi(\mathcal{L})$  for  $i = 1, \dots, N$   $\triangleright \Phi(\mathcal{L}) = \text{fundamental domain}$   
 Let  $Z = \{(x_1, y_1), \dots, (x_N, y_N)\}$   
 $R \leftarrow R_0$
2. **while**  $R > 6$  **do**  
 Find  $J \subseteq I$  and  $\varphi : I \rightarrow J$ ,  $|J| \leq 5^n$  such that  $\forall i \in I : \|y_i - y_{\varphi(i)}\| \leq \frac{R}{2}$   
 $I \leftarrow I \setminus J$   
 $Z \leftarrow \{(x_i, y_i - (y_{\varphi(i)} - x_{\varphi(i)})) \mid i \in I\}$   
 $R \leftarrow \frac{R}{2} + 2$
3. For any two pairs  $(x_i, y_i), (x_j, y_j)$  in  $Z$  consider the difference  $(y_i - x_i) - (y_j - x_j)$  and output the shortest nonzero vector among these differences

**Remark.** *The condition  $R > 6$  ensures we do not accidentally reduce all vectors to zero. Since we assume  $\lambda_1 \geq 2$ , if we subtracted vectors to get a result with norm  $< 2$ , it would have to be the zero vector. By keeping the radius bounds appropriately, we ensure collisions  $(x_i - x_j)$  yield non-zero short vectors. The complexity is dominated by the number of points  $N \approx 5^n$ .*

## 12 Algorithm for SVP (Analysis)

In this lecture, we analyze the correctness and running time of the AKS algorithm presented previously.

### Algorithm Properties (Lemmas 1–5).

**Lemma** (#1, Initialization Properties). *Let  $R_0 = n \max_i \|b_i\|$ . Suppose  $x \in B(0, 2)$  is chosen uniformly at random and  $y \equiv x \pmod{\Phi(B)}$ , where  $\Phi(B)$  is the fundamental parallelepiped. Then  $x - y \in \mathcal{L}$  and  $\|y\| \leq R_0$ .*

*Proof.* Recall that  $\Phi(B) = \{\sum \gamma_i b_i \mid \gamma_i \in [0, 1)\}$  and  $\mathcal{L} = \{\sum z_i b_i \mid z_i \in \mathbb{Z}\}$ . For any  $x \in B(0, 2)$ , we can write  $x = \sum \gamma_i b_i$  with  $\gamma_i \in \mathbb{R}$ . The vector  $y = x \bmod \Phi(B)$  is defined as  $\sum (\gamma_i - \lfloor \gamma_i \rfloor) b_i$ . Since  $\lfloor \gamma_i \rfloor \in \mathbb{Z}$ , the difference  $x - y = \sum \lfloor \gamma_i \rfloor b_i$  is an integer linear combination of basis vectors, hence  $x - y \in \mathcal{L}$ . For the norm bound, since  $\gamma_i - \lfloor \gamma_i \rfloor \in [0, 1)$ , we have:

$$\|y\| = \left\| \sum (\gamma_i - \lfloor \gamma_i \rfloor) b_i \right\| \leq \sum |\gamma_i - \lfloor \gamma_i \rfloor| \|b_i\| \leq \sum \|b_i\| \leq n \max_i \|b_i\| = R_0.$$

□

**Lemma** (#2, Step 2 Norm Reduction). *After one iteration of [Step 2],  $\|y_i\| \leq R/2 + 2$ .*

*Proof.* Using the triangle inequality:

$$\|y_i^{new}\| = \|y_i - (y_{\varphi(i)} - x_{\varphi(i)})\| \leq \|y_i - y_{\varphi(i)}\| + \|x_{\varphi(i)}\| \leq \frac{R}{2} + 2.$$

Note that  $(y_i - x_i) - (y_{\varphi(i)} - x_{\varphi(i)}) \in \mathcal{L}$  is maintained, so the lattice invariant holds. □

**Lemma** (#3, Number of Iterations). *The number of iterations of the WHILE loop in [Step 2] is at most  $2 \log R_0$ .*

*Proof.* Let  $R_k$  be the value of  $R$  at the start of the  $k$ -th iteration.  $R_1 = R_0$ . The recurrence relation is  $R_{k+1} = \frac{R_k}{2} + 2$ . Expanding this:

$$R_k = \frac{R_0}{2^{k-1}} + 2 + \sum_{j=0}^{k-3} \frac{1}{2^j}$$

We stop when  $R_k \leq 6$ . If we set  $k = \lceil \log R_0 \rceil + 2$ , then

$$R_k = \frac{R_0}{2^{\lceil \log R_0 \rceil}} + 2 + \sum_{j=0}^{\lceil \log R_0 \rceil - 2} \frac{1}{2^j} \leq 1 + 2 + 2 = 5 < 6.$$

so terminates the loop. Thus, the number of iterations is bounded by  $O(\log R_0) \leq 2 \log R_0$ . □

**Lemma** (#4, Time Complexity). *The AKS algorithm runs in  $2^{O(n)} \cdot \text{poly}(\text{size of input})$ .*

*Proof.* Let  $T = \log R_0$  be roughly the input size. Step 1 involves generating  $N = 2^{8n} \log R_0$  points, which takes  $2^{O(n)} \text{poly}(T)$ . The Sieve procedure runs in polynomial time in  $N$ , i.e.,  $(2^{O(n)})^c = 2^{O(n)}$ . By Lemma 3, the loop runs at most  $2T$  times. Thus, the total time is dominated by  $N \cdot \text{iterations} \approx 2^{O(n)} \text{poly}(T)$ . □

**Lemma** (#5, Population Size). *At the end of [Step 2], the set  $Z$  contains at least  $(2^{8n} - 2 \cdot 5^n) \log R_0$  pairs  $(x_i, y_i)$ . Each pair yields a lattice vector  $u = x_i - y_i$  with  $\|u\| \leq 8$ .*

*Proof.* Initially,  $|Z| = 2^{8n} \log R_0$ . In each iteration of [Step 2], the sieving removes at most  $|J| \leq 5^n$  indices (the centers). Since there are at most  $2 \log R_0$  iterations, the total number of removed pairs is at most  $5^n \cdot 2 \log R_0$ . The remaining size is at least  $(2^{8n} - 2 \cdot 5^n) \log R_0$ . In the final state,  $R \leq 6$ . Since  $\|y_i\| \leq R \leq 6$ ,  $\|x_i\| \leq 2$ , and by Lemma 1  $y_i - x_i \in \mathcal{L}$ , we get by triangular inequality:

$$\|x_i - y_i\| \leq \|x_i\| + \|y_i\| \leq 2 + 6 = 8.$$

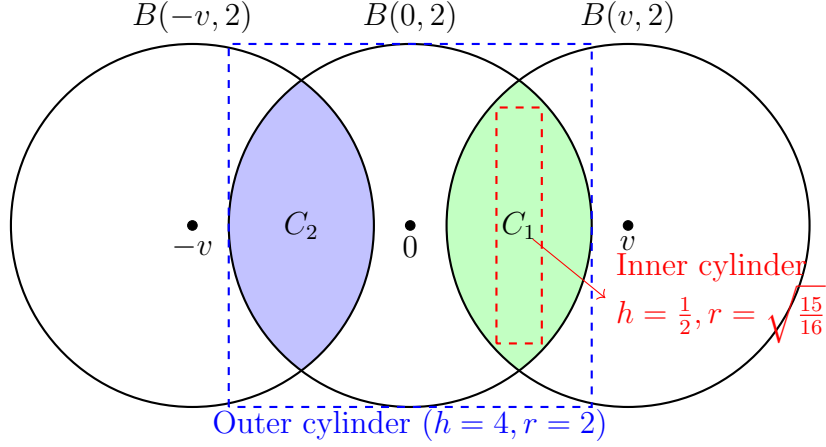
□

## Correctness of AKS (with high probability).

Let  $v$  be a shortest non-zero vector in  $\mathcal{L}$ , so  $\|v\| = \lambda_1 \in [2, 3]$ . Define the regions of "good" samples:

$$C_1 = B(0, 2) \cap B(v, 2) \quad \text{and} \quad C_2 = B(0, 2) \cap B(-v, 2).$$

**Definition** (Green point). A point  $x_i$  is called green if  $x_i \in C_1 \cup C_2$ .



**Lemma** (#6, Volume Bound).

$$\frac{\text{vol}(C_1)}{\text{vol}(B(0, 2))} = \frac{\text{vol}(C_2)}{\text{vol}(B(0, 2))} \geq 2^{-2n}.$$

*Proof.* By symmetry,  $\text{vol}(C_1) = \text{vol}(C_2)$ , so the first equality holds. We lower bound  $\text{vol}(C_1)$  by inscribing a cylinder of height  $1/2$  (along the  $v$  direction) and radius  $r = \sqrt{\frac{15}{16}}$  centered at  $v/2$  completely inside  $C_1$ . Thus:

$$\text{vol}(C_1) \geq \frac{1}{2} \cdot \text{vol}(B_{n-1}(0, \sqrt{15/16})).$$

On the other hand,  $B(0, 2)$  is contained in a cylinder of height 4 (from -2 to 2) and radius 2.

$$\text{vol}(B(0, 2)) \leq 4 \cdot \text{vol}(B_{n-1}(0, 2)).$$

Taking the ratio:

$$\frac{\text{vol}(C_1)}{\text{vol}(B(0, 2))} \geq \frac{\frac{1}{2} \text{vol}(B_{n-1}(0, \sqrt{15/16}))}{4 \text{vol}(B_{n-1}(0, 2))} = \frac{1}{8} \left( \frac{\sqrt{15/16}}{2} \right)^{n-1} = \frac{1}{8} \left( \frac{15}{64} \right)^{\frac{n-1}{2}}.$$

For large  $n$ ,  $(15/64) \approx 1/4$ , so roughly  $(1/4)^{n/2} = (1/2)^n$ . The exact bound derived in the lecture notes is  $\geq 2^{-2n}$ .  $\square$

**Lemma** (#7, Number of Green Points). With high probability, there are at least  $2^{6n-1} \log R_0$  green points in the initial set  $X$ .

*Proof.* Let  $p = \Pr[x_i \text{ is green}]$ . Since  $C_1$  and  $C_2$  are disjoint (for  $\|v\| \geq 2$ ),

$$p = \frac{\text{vol}(C_1 \cup C_2)}{\text{vol}(B(0, 2))} = \frac{2 \text{vol}(C_1)}{\text{vol}(B(0, 2))} \geq 2 \cdot 2^{-2n} > 2^{-2n}.$$

Let  $K$  be the number of green points in the sample size  $N = 2^{8n} \log R_0$ . The expected number is  $\mathbb{E}[K] = pN \geq 2^{-2n} \cdot 2^{8n} \log R_0 = 2^{6n} \log R_0$ . We use Chebyshev's inequality to bound the probability of having too few points. The variance is  $\text{Var}(K) = Np(1-p) < Np = \mathbb{E}[K]$ .

$$\Pr[K < \frac{1}{2} \mathbb{E}[K]] \leq \frac{\text{Var}(K)}{(\mathbb{E}[K]/2)^2} < \frac{\mathbb{E}[K]}{\mathbb{E}[K]^2/4} = \frac{4}{\mathbb{E}[K]}.$$

Since  $\mathbb{E}[K]$  is exponential in  $n$ , this probability is exponentially small. Thus, with high probability,  $K \geq \frac{1}{2} 2^{6n} \log R_0 = 2^{6n-1} \log R_0$ .  $\square$

**Conclusion of Correctness:** We end up with  $> 2^{6n}$  green points. Since the number of lattice points in  $B(0, 8)$  is bounded by  $2^{4n}$  (Claim 5 in notes), by Pigeonhole Principle, there must be many collisions  $y_i - x_i = y_j - x_j$ , allowing us to recover the shortest vector  $v$  as their difference.



## 13 Hermite Normal Form

**Definition (HNF).** We say that a matrix  $A = (a_{i,j}) \in M_{n,m}(\mathbb{Z})$  is in Hermite normal form (HNF) if there exists  $r \leq m$  and a strictly increasing map  $f$  from  $[r+1, m] \rightarrow [1, n]$  satisfying the following properties:

- (1) For  $r+1 \leq j \leq m$ ,  $a_{f(j),j} \geq 1$ ,  $a_{i,j} = 0$  if  $i > f(j)$  and  $0 \leq a_{f(k),j} < a_{f(k),k}$  if  $k < j$ .
- (2) The first  $r$  columns of  $A$  are equal to 0.

**Remark.** In the important special case where  $m = n$  and  $f(k) = k$  (or equivalently  $\det(M) \neq 0$ ),  $A$  is in HNF if it satisfies the following conditions:

1.  $A$  is an upper triangular matrix, i.e.,  $a_{i,j} = 0$  if  $i > j$ .
2. For every  $i$ , we have  $a_{i,i} > 0$ .
3. For every  $j > i$  we have  $0 \leq a_{i,j} < a_{i,i}$ .

**Theorem.** Let  $A \in M_{n,m}(\mathbb{Z})$ . Then there exists a unique matrix  $B \in M_{n,m}(\mathbb{Z})$  in HNF of the form  $B = AU$  with  $U \in GL_n(\mathbb{Z})$ .

The matrix  $B$  is called the Hermite normal form of  $A$ .

**Algorithm (Hermite Normal Form).**

Given an  $m \times n$  matrix  $A$  with integer coefficients  $(a_{i,j})$ , this algorithm finds the Hermite normal form  $W$  of  $A$ . We write  $W_j$  for the columns of  $W$  and  $A_j$  for the columns of  $A$ .

1. **[Initialize]** Set  $i \leftarrow m$ ,  $k \leftarrow n$ ,  $l = 1$  if  $m \leq n$ ,  $l = m - n + 1$  if  $m > n$ .
2. **[Row finished?]** If all the  $a_{i,j}$  with  $j < k$  are zero, then if  $a_{i,k} < 0$  replace column  $A_k$  by  $-A_k$  and go to step 5.
3. **[Choose non-zero entry]** Pick among the non-zero  $a_{i,j}$  for  $j \leq k$  one with the smallest absolute value, say  $a_{i,j_0}$ . Then if  $j_0 < k$ , exchange column  $A_k$  with column  $A_{j_0}$ . In addition, if  $a_{i,k} < 0$  replace column  $A_k$  by  $-A_k$ . Set  $b \leftarrow a_{i,k}$ .
4. **[Reduce]** For  $j = 1, \dots, k-1$  do the following: set  $q \leftarrow \lfloor a_{i,j}/b \rfloor$ , and  $A_j \leftarrow A_j - qA_k$ . Then go to step 2.
5. **[Final reductions]** Set  $b \leftarrow a_{i,k}$ . If  $b = 0$ , set  $k \leftarrow k+1$  and go to step 6. Otherwise, for  $j > k$  do the following: set  $q \leftarrow \lfloor a_{i,j}/b \rfloor$ , and  $A_j \leftarrow A_j - qA_k$ .
6. **[Finished?]** If  $i = l$  then for  $j = 1, \dots, n-k+1$  set  $W_j \leftarrow A_{j+k-1}$  and terminate the algorithm. Otherwise, set  $i \leftarrow i-1$ ,  $k \leftarrow k-1$  and go to step 2.

# 14 Structure Theorem and Smith Normal Form

## Free Abelian Groups.

**Definition** (Free Abelian Group). A  $\mathbb{Z}$ -module (Abelian group)  $V$  is called finitely generated if there exists a finite set  $(v_i)_{1 \leq i \leq k}$  of elements of  $V$  such that any element of  $V$  can be expressed as a linear combination of the  $v_i$  with integral coefficients.

$V$  is called a free  $\mathbb{Z}$ -module (or free abelian group) if  $V \simeq \mathbb{Z}^n$  for some  $n$ . The set of generators is called a free basis.

**Example** (Lattices). Free  $\mathbb{Z}$ -modules of finite rank are often denoted by  $L$  and called lattices. We usually consider  $L$  to be a submodule of some  $\mathbb{Z}^m$ .

**Proposition.** Every finitely generated free abelian group is isomorphic to the group  $\mathbb{Z}^n$  for some  $n$ .

**Proposition.**  $\mathbb{Z}^n$  and  $\mathbb{Z}^m$  are isomorphic  $\iff n = m$ .

*Proof.* First implication “ $\Leftarrow$ ” is trivial.

Second “ $\Rightarrow$ ”: Let  $\phi : \mathbb{Z}^n \rightarrow \mathbb{Z}^m$  be an isomorphism. Then it induces an isomorphism of vector spaces over  $\mathbb{Q}$ :  $\mathbb{Z}^n \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{Z}^m \otimes_{\mathbb{Z}} \mathbb{Q}$ , which implies  $\mathbb{Q}^n \cong \mathbb{Q}^m$ . Since the dimension of a vector space is invariant,  $n = m$ .  $\square$

**Theorem** (Structure theorem). Let  $V$  be a finitely generated module over a PID (e.g.,  $\mathbb{Z}$ ). Then:

- (1)  $V \simeq V_{tors} \times \mathbb{Z}^n$  where  $V_{tors}$  is the torsion subgroup.
- (2) If  $V$  is free, any submodule is free of rank  $\leq$  rank of  $V$ .
- (3) If  $V$  is a finite  $R$ -module (rank 0),  $V \simeq R^n/L$  for some submodule  $L$  of  $R^n$ .

## Smith Normal Form (SNF)

**Definition** (SNF). A matrix  $B = (b_{i,j}) \in M_{n,n}(\mathbb{Z})$  is in Smith Normal Form if it is a diagonal matrix with nonnegative integer coefficients, such that  $b_{i+1,i+1} | b_{i,i}$ .

**Theorem.** Let  $A \in M_{nn}(\mathbb{Z})$  with  $\det(A) \neq 0$ . Then there exists a unique matrix  $B$  in SNF such that  $B = VAU$  with  $U, V \in GL_n(\mathbb{Z})$ .

**Definition.** We say that integer  $d_i$  is the  $i$ -th elementary divisor of  $A$ , where

$$A = V \begin{pmatrix} d_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_n \end{pmatrix} U.$$

**Theorem** (Elementary Divisors Theorem). Let  $V$  be a free  $\mathbb{Z}$ -module of rank  $n$ , and  $V'$  a submodule of the same rank. Then there exists positive integers  $d_1, \dots, d_n$  such that:

- (1)  $d_{i+1} | d_i$  for all  $i \in \{1, \dots, n\}$ .
- (2) there exists a  $\mathbb{Z}$ -basis  $v_1, \dots, v_n$  of  $V'$ , such that  $d_1 v_1, \dots, d_n v_n$  is a basis of  $V$ .
- (3) a  $\mathbb{Z}$ -modules isomorphism  $V'/V \simeq \mathbb{Z}_{d_1\mathbb{Z}} \oplus \dots \oplus \mathbb{Z}_{d_n\mathbb{Z}}$  and  $[V' : V] = d_1 \dots d_n$  and  $d_1$  is an exponent of  $V'/V$ .

We calculate the SNF of an  $n \times n$  non-singular integral matrix  $A$ . All reductions modulo  $R$  are taken in the interval  $] - R/2, R/2]$ .

**Algorithm** (Smith Normal Form).

*Input:*  $n \times n$  non-singular integral matrix  $A = (a_{i,j})$ .

*Output:* Diagonal elements  $d_i$  such that  $d_{i+1} | d_i$ .

1. **[Initialize i]** Set  $i \leftarrow n$ ,  $R \leftarrow |\det(A)|$ . If  $n = 1$ , output  $d_1 \leftarrow R$  and terminate.
2. **[Initialize j for row reduction]** Set  $j \leftarrow i$ ,  $c \leftarrow 0$ .
3. **[Check zero]** If  $j = 1$  go to step 5. Otherwise, set  $j \leftarrow j - 1$ . If  $a_{i,j} = 0$  go to step 3.
4. **[Euclidean step]** Using Euclid's extended algorithm, compute  $(u, v, d)$  such that  $ua_{i,i} + va_{i,j} = d = \gcd(a_{i,i}, a_{i,j})$ , with  $u$  and  $v$  minimal. Then set  $B \leftarrow uA_i + vA_j$ ,  $A_j \leftarrow ((a_{i,i}/d)A_j - (a_{i,j}/d)A_i) \pmod{R}$ ,  $A_i \leftarrow B \pmod{R}$  and go to step 3.
5. **[Initialize j for column reduction]** Set  $j \leftarrow i$ .
6. **[Check zero]** If  $j = 1$  go to step 8. Otherwise, set  $j \leftarrow j - 1$ , and if  $a_{j,i} = 0$  go to step 6.
7. **[Euclidean step]** Using Euclid's extended algorithm, compute  $(u, v, d)$  such that  $ua_{i,i} + va_{j,i} = d = \gcd(a_{i,i}, a_{j,i})$ , with  $u$  and  $v$  minimal. Then set  $B \leftarrow uA'_i + vA'_j$ ,  $A'_j \leftarrow ((a_{i,i}/d)A'_j - (a_{j,i}/d)A'_i) \pmod{R}$ ,  $A'_i \leftarrow B \pmod{R}$ ,  $c \leftarrow c + 1$  and go to step 6.
8. **[Repeat stage i?]** If  $c > 0$  go to step 2.
9. **[Check the rest of the matrix]** Set  $b \leftarrow a_{i,i}$ . For  $1 \leq k, l < i$  check whether  $b|a_{k,l}$ . As soon as some coefficient  $a_{k,l}$  is not divisible by  $b$ , set  $A'_i \leftarrow A'_i + A'_k$  and go to step 2.
10. **[Next stage]** Output  $d_i = \gcd(a_{i,i}, R)$  and set  $R \leftarrow R/d_i$ . If  $i = 2$ , output  $d_1 = \gcd(a_{1,1}, R)$  and terminate. Otherwise, set  $i \leftarrow i - 1$  and go to step 2.

## Application to Finite Groups.

**Theorem.** *Let  $G$  be a finitely generated abelian group (finite  $\mathbb{Z}$ -module). Then there exist finitely generated free abelian groups  $H, F$  such that  $G$  is isomorphic to  $H/F$ .*

*Proof.* Let  $g_1, \dots, g_n$  be a set of generators for  $G$ . Let  $H = \mathbb{Z}^n$  be the free abelian group on  $n$  generators  $e_1, \dots, e_n$ . Define a homomorphism  $\phi : H \rightarrow G$  by  $\phi(e_i) = g_i$ . This is surjective by definition. Let  $F = \ker(\phi)$ . By the First Isomorphism Theorem,  $G \cong H/F$ . Since  $H$  is free, the submodule  $F$  is also free (by Theorem).  $\square$

## Sources

- General – Lattices in Computer Science by Oded Regev
- Lecture 1 – Micciancio–Goldwasser - Basics, Discrete subgroups – Neukirch [pp. 24–25]
- Lecture 2 – Micciancio–Goldwasser - Basics
- Lecture 3 – Micciancio–Goldwasser - Basics
- Lecture 5 – Micciancio–Goldwasser [p. 32–34]
- Lecture 6 – Micciancio–Goldwasser [p. 34–36]
- Lecture 8 – Regev, Coppersmith
- Lecture 9 – Regev, Babai
- Lecture 10 – Regev, Micciancio–Goldwasser [p. 45–49]
- Lecture 11 – Regev
- Lecture 12 – Regev
- Lecture 13 – Cohen [p. 66]
- Lecture 14 – Cohen [p. 65, 76]